# Aging populations and strategies for public debt in Europe*

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September 20, 2001
**Perliminary and incomplete version**

**Abstract**

In the coming 30 to 40 years, the share of people in working age is predicted to fall significantly in all European countries. Considering this reduction in the relative size of the labor force, it may be wise to start adjusting public policy already now. I examine 12 European countries and find that for all these countries, the optimal policy is to start reducing public debt immediately and continuously until sometime between year 2030 and 2045. The implied optimal budget surplus on average over these years varies between one and four percent of output annually.

**Keywords:** public policy, public debt, optimal taxation, demographics
**JEL Classification:** E62, H21, H60, J18

*I thank Torben M. Andersen and Per Molander for helpful suggestions. This work was initially produced for a project on 'The scope and size of the public sector' funded by SNS (Center for Business and Policy Studies). I also thank the Wallander and Hedelius Foundation at Svenska Handelsbanken for generous funding.

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# 1 Introduction

During the final decades of the previous century, most European countries accumulated large, and in peace time unprecedented, public debts. Interest payments on the debts now constitute an important fiscal burden. Today these countries also face another challenge to the public budget. In the coming 30 to 40 years, the share of people in working age is predicted to fall significantly. Considering this reduction in the relative size of the labor force, it may be wise to start adjusting public policy already now. In particular, fiscal constraints may become less tight in the future if public debts are reduced today. This paper explores what paths for the public debt and the budget surplus are optimal in the face of aging populations.

Twelve European countries are included in the study. For all these countries, the optimal policy is to start reducing public debt immediately and continuously until sometime between year 2030 and 2045. The optimal average budget surplus varies between one and four percent of output annually for these countries. The calculations show that for most countries the welfare loss of pursuing a balanced-budget policy instead of the optimal debt policy would be modest but not negligible.

## 1.1 The demographic change

Figures 1 to 3 show the historical and predicted demographic development in the countries considered.[1] Figure 1 clearly displays that the share of workers in the population will fall during the coming 30 years. In Sweden, for example, this share is predicted to fall from 59 percent in year 2001 to 52 percent in year 2035. The figure also shows that the population aging is predicted to be similar in all countries, but that the demographic impact will be largest in Southern Europe.

Figure 2 shows that the share of old in the population has increased and is predicted to continue to increase. The pattern is again similar for all countries, but most remarkable in Southern Europe. In Sweden, the fraction of retirees is predicted to increase from 19 percent today to 29 percent in year 2035. Figure 3 shows that the increased dependency burden from population aging will be offset by a reduced fraction of children in the population. The fraction of children will fall from 24 to 21 percent in Sweden.

Calculations in Domeij and Floden (2001) show that, at least for Sweden, increased life expectancy and falling birth rates contribute to population aging approximately to the same extent.

## 1.2 Impact on public budget, saving, and production

Population aging can influence the overall economy and, in particular, the public budget in several important ways. Before developing these issues further, let us consider a simple experiment conducted to demonstrate that the changing age structure is likely to be quantitatively important. The experiment is based on the demographic forecasts for Sweden. Assume that future government policy will be identical to the average policy in the recent past, and that everyone in working age supplies one unit of labor. Further, assume that public transfers and public consumption per capita is constant and equally divided between different age groups. Figure 4 shows the resulting development of tax revenues and public spending (not including interest

[1] People aged 20 to 64 are assumed to be in working age, while people under 20 are 'young' and people above 64 are 'old' or 'retired'. Data are from the United Nations, 1998.

payments) over time. The experiment predicts that the primary surplus will deteriorate by five percentage points in the coming 40 years. Calculations by the OECD (1998, p. 33) result in a quantitatively similar effect on the public budget.[2]

The real response to the demographic change will be more complicated than what the above experiment indicates. First, individuals may change their behavior in response to changing demographic structures. For example, Bohn (1999) and Domeij and Floden (2001) show that if people live longer, incentives to save and work as young may be strengthened since people then want to accumulate more reserves for their old age. Moreover, if the population aging implies that future pension benefits will be less generous, people will save more as young to compensate for that reduction. Since people save more, the capital stock may increase and productivity per worker will increase. The increase in savings can therefore mitigate the demographic impact on production. Domeij and Floden demonstrate that these offsetting effects are most likely to occur in countries with little distortive taxation. Consequently, countries with a large public sector are least likely to be helped by the individual response to population aging.

Second, the interest rate and wage rate are likely to change during the demographic transition. As the number of workers falls, capital will become less productive and the interest rate will fall. Cutler et al. (1990) and Elmendorf and Scheiner (2000a, 2000b) therefore argue that the optimal response to an increased dependency burden is to reduce the national savings rate.

Third, as the experiment above indicates, maintaining an unaltered public policy will not be sustainable, and possibly not desirable. Expenses must be cut or revenue increased. The model used below is developed to examine how revenue should vary if expenses are held fixed at the current level. To some extent, the tax increases can equivalently be interpreted as expenditure cuts. However, Domeij and Floden (2001) show that savings responses to tax increases and benefit cuts can be very different, at least if individuals are not altruistic to other household members. For example, people will increase savings today if they expect future pension benefits to be low, while an increase in future social security taxes does not directly affect today's workers.

Fourth, the experiment ignores that worker productivity (wages and labor-market participation) and consumption needs may vary with age and time. In the model spelled out below, effective labor supply is adjusted for age-specific productivity and participation rates. Throughout in the study, public spending and public transfers per capita are assumed to be held constant at the present level.[3] Inferring the present level from the data is, however, not unproblematic. As the benchmark, all public outlays are assumed to be uniformly divided between age groups. An alternative parameterization of the model attempts to assign age-specific values to public outlays.

Health care costs account for a substantial part of public outlays. Since the elderly consume more health care than other groups, one could expect these costs to increase as the population grows older. But, as life expectancy increases people of a certain age will become healthier and thus demand less care. Several studies indeed indicate that health care costs are more closely related to the remaining life time than to age.[4] The model specification with age-specific public expenses may therefore exaggerate the future increase in public outlays.

[2] The OECD calculations also include some (not explicit) predictions of changes in policy and behaviour. The OECD calculations for Belgium, Canada, France, Germany, and the Netherlands are quantitatively similar. The demographic impact is predicted to be slightly smaller in the U.S., negligible in the U.K., and most severe in Japan.

[3] There is no explicit productivity growth in the model, so 'constant' should be interpreted as 'constant relative to productivity'.

[4] See for example the discussion in chapter 4 in Batljan and Lagergren (2000).

## 2 The model

Consider an economy populated by a large number of identical and infinitely lived households of unit mass. A fraction $\eta$ of household members are active in the labor market and have one unit of time to dispose of. Members of the household maximize their joint utility described by

$$\sum_{t=0}^{\infty} \beta^t U\left(c_t^a, c_t^i, h_t, g, \eta_t\right) \tag{1}$$

where $U$ is the instantaneous utility, $c^a$ and $c^i$ are consumption per active and inactive household member, respectively, $h$ is labor supply per worker, and $g$ is public consumption.

Let $\nu$ denote the efficiency of a worker, and let $H = \eta\nu h$ denote a household's total labor supply in efficiency units.[5] The household budget constraint is then

$$a_{t+1} = \hat{R}_t a_t + \left(1 - \tau_t^h\right) w_t H_t + b - (1 + \tau^c)\left[\eta_t c_t^a + (1 - \eta_t) c_t^i\right] \tag{2}$$

where $a_{t+1}$ is savings from period $t$ to period $t+1$, $\hat{R}_t = 1 + (1 - \tau^a) r_t$, $\tau^a$ is the tax rate that applies to (domestic) households' asset returns, $\tau^h$ is the labor-income tax rate, $w$ is the wage rate, $b$ is a lump sum transfer from the government, and $\tau^c$ is the consumption tax.

The budget constraint can be rewritten as a life-time constraint,

$$\sum p_t \left[(1 + \tau^c)\left(\eta_t c_t^a + (1 - \eta_t) c_t^i\right) - \left(1 - \tau_t^h\right) w_t H_t - b\right] = \hat{R}_0 a_0 \tag{3}$$

where $p_t / p_{t-1} = 1/\hat{R}_t$. The household's first order conditions are

$$\frac{U_{1t}}{\eta_t} = \frac{U_{2t}}{1 - \eta_t} \tag{4}$$

$$\frac{U_{3t}}{U_{1t}} = \frac{-\left(1 - \tau_t^h\right) \nu_t w_t}{1 + \tau^c} \tag{5}$$

$$\beta^t U_{1t} = \lambda p_t \eta_t (1 + \tau^c) \tag{6}$$

where $\lambda$ is the Lagrange multiplier on the budget constraint. If $p_0$ is normalized to unity, the household budget constraint reduces to

$$\sum \beta^t \left[U_{1t}\left(c_t^a - \frac{b}{\eta_t (1 + \tau^c)}\right) + U_{2t} c_t^i + U_{3t} h_t\right] = \frac{U_{10} \hat{R}_0 a_0}{\eta_0 (1 + \tau^c)} \tag{7}$$

A large number of competitive firms maximize profits,

$$\max F(k, \eta\nu h) - w\eta\nu h - \left[(1 + \tau^\pi) r + \delta\right] k$$

where $F$ is the production function, $k$ is the capital stock, $\tau^\pi$ is the tax on firm profits, and $\delta$ is the depreciation rate of capital. Competition ensures that

$$(1 + \tau^\pi) r_t = F_{kt} - \delta \tag{8}$$

$$w_t = F_{ht} \tag{9}$$

[5] Productivity and labor-market participation varies with age. Thus, $\nu$ captures variations in efficiency due to variations in the age composition of the labor force.

$F_k$ and $F_h$ are the marginal products of capital and labor.

Let $R_t = 1 + r_t$. The government's budget constraint is then

$$d_{t+1} = R_t d_t + g + b - \tau_t^h w_t H_t - \tau^a r_t a_t - \tau^\pi r_t k_t - \tau^c \left[\eta_t c_t^a + (1-\eta_t) c_t^i\right] \tag{10}$$

By substituting the household budget constraint (2) into (10), the government's budget constraint can be rewritten as

$$\sum q_t \left(g + \eta_t c_t^a + (1-\eta_t) c_t^i - \tau^\pi r_t k_t - w_t H_t\right) = R_0 (a_0 - d_0) \tag{11}$$

where $q_t/q_{t-1} = 1/R_t$, and $q_0$ is normalized to unity.

### 2.1 The world economy

A small open economy will take factor prices for given. These prices are determined on the world market, and I assume that policy in the world economy will balance the public budget in each period.[6]

The world economy is thus characterized by a sequence of tax rates, $\left\{\hat{\tau}_t^h\right\}$, that fulfills the budget constraint

$$0 = r_t d + g_t + b_t - \hat{\tau}^h w_t H_t - \tau^a r_t a_t - \tau^\pi r_t k_t - \tau^c \left[\eta_t c_t^a + (1-\eta_t) c_t^i\right] \tag{12}$$

for all periods $t$. In this budget constraint, the levels of debt, $d$, public expenditure, $g$, and transfers $b$, are exogenous, as is the demographic development, $\eta$, and the tax rates on firm profits, $\tau^\pi$, and consumption, $\tau^c$. In a closed economy, total household asset holdings must equal the sum of the capital stock and total government debt,

$$a_t = k_t + d$$

The interest rate, wage rate, and capital stock are determined by this restriction in combination with (8) and (9). Household decisions for $a_{t+1}$, $h_t$, $c_t^a$, and $c_t^i$ are determined by the first order conditions (4) - (6) in combination with the household budget constraint (2). The initial capital stock, $k_0$, and initial household assets, $a_0$, are given.

### 2.2 Optimal debt policy in a small open economy

The interest rate path $\{r_t\}$ is exogenous to the small open economy. Capital can move freely between countries but labor is immobile. Competition still ensures that interest rates and wages equal the marginal products of capital and labor. Equations (8) and (9) must therefore hold also in the small open economy.

A feasible government policy is a sequence of tax rates $\left\{\tau_t^h\right\}$ fulfilling the budget constraint and a transversality condition. To find the optimal policy, it is convenient to reformulate the government's optimization problem as a Ramsey allocation problem where the government chooses sequences of consumption and labor supply under the additional constraint that these sequences are consistent with household optimization. For more on the Ramsey allocation problem, see Chari and Kehoe (1999) and Atkeson, Chari, and Kehoe (1999).[7]

[6] This assumption is arbitrary and will not be consistent with the implications of optimal public policy. As an alternative, I also consider the implications of assuming that future prices will be constant at the present levels.

[7] Their sections on open economy models are particularly relevant.

The Ramsey allocation problem is thus

$$\max_{\{c_t^a, c_t^i, h_t\}} \sum \beta^t U\left(c_t^a, c_t^i, h_t, g, \eta_t\right)$$

subject to the household and government budget constraints, (7) and (11), and household optimization, (4) and (6). Note that one of the household optimization conditions, equation (5), is used to solve for the labor tax as a function of allocations.

Let $\rho_1$, $\rho_2, \rho_{3t}$, and $\rho_{4t}$ be the Lagrange multipliers associated with (7), (11), (4) and (6), respectively. Further, let

$$\begin{aligned} W_t &= W\left(c_t^a, c_t^i, h_t, g, \eta_t, \rho_1\right) \\ &= U\left(c_t^a, c_t^i, h_t, g, \eta_t\right) + \rho_1\left[U_{1t}\left(c_t^a - \frac{b}{\eta_t\left(1+\tau^c\right)}\right) + U_{2t}c_t^i + U_{3t}h_t\right] \end{aligned}$$

and let $\kappa = k/y$. The Lagrangian to the Ramsey problem is then

$$\begin{aligned} \mathcal{L} &= \sum_{t=0}^{\infty} \beta^t W_t + \\ & \rho_2 \sum_{t=0}^{\infty} q_t\left[g + \eta_t c_t^a + \left(1-\eta_t\right)c_t^i - \left(\tau^\pi r_t \kappa^{\frac{1}{1-\theta}} + \left(1-\theta\right)\kappa^{\frac{\theta}{1-\theta}}\right)\eta_t v_t h_t\right] + \\ & \sum_{t=0}^{\infty} \rho_{3t}\left[\left(1-\eta_t\right)U_{1t} - \eta_t U_{2t}\right] + \sum_{t=0}^{\infty} \rho_{4t}\left[\beta \hat{R}_{t+1} U_{1t+1}/\eta_{t+1} - U_{1t}/\eta_t\right] - \\ & \rho_1 U_{10}\hat{R}_0 a_0/\eta_0 - \rho_2 R_0\left(a_0 - d_0\right) \end{aligned}$$

and the first order conditions with respect to $c_0^a$, $c_0^i$, and $h_0$ (assuming utility of $c^i$ is separable from $c^a$ and $h$) are

$$\begin{aligned} 0 &= W_{10} - \rho_1 U_{110}\hat{R}_0 a_0/\eta_0 + \rho_2\eta_0 + \rho_{30}\left(1-\eta_0\right)U_{110} - \rho_{40}U_{110}/\eta_0 \\ 0 &= W_{20} + \rho_2\left(1-\eta_0\right) - \rho_{30}\eta_0 U_{220} \\ 0 &= W_{30} - \rho_1 U_{130}\hat{R}_0 a_0/\eta_0 - \rho_2\eta_0\nu_0\left(\tau^\pi r_t \kappa^{\frac{1}{1-\theta}} + \left(1-\theta\right)\kappa^{\frac{\theta}{1-\theta}}\right) \\ & \quad + \rho_{30}\left(1-\eta_0\right)U_{130} - \rho_{40}U_{130}/\eta_0 \end{aligned}$$

The first order conditions with respect to. $c_t^a$, $c_t^i$, and $h_t$, for $t > 0$ are

$$\begin{aligned} 0 &= \beta^t W_{1t} + \rho_2 q_t \eta_t + \rho_{3t}\left(1-\eta_t\right)U_{11t} - \rho_{4t}U_{11t}/\eta_t + \rho_{4t-1}\beta\hat{R}_t U_{11t}/\eta_t \\ 0 &= \beta^t W_{2t} + \rho_2 q_t\left(1-\eta_t\right) - \rho_{3t}\eta_t U_{22t} \\ 0 &= \beta^t W_{3t} - \rho_2 q_t \eta_t \nu_t\left(\tau^\pi r_t \kappa^{\frac{1}{1-\theta}} + \left(1-\theta\right)\kappa^{\frac{\theta}{1-\theta}}\right) + \rho_{3t}\left(1-\eta_t\right)U_{13t} \\ & \quad - \rho_{4t}U_{13t}/\eta_t + \rho_{4t-1}\beta\hat{R}_t U_{13t}/\eta_t \end{aligned}$$

Clearly, there can only be a stationary equilibrium if $\beta^t = q_t$ in the long run. In steady state, therefore, $\beta R = 1$.

## 3 Calibration

The utility function is assumed to be

$$U\left(c^a, c^i, h, g, \eta\right) = \eta \frac{\left(c^a\right)^{1-\mu}}{1-\mu} \exp\left[-\zeta\left(1-\mu\right) h^{1+1/\gamma}\right] + \left(1-\eta\right) \frac{\left(c^i\right)^{1-\mu}}{1-\mu} + v\left(g\right)$$

where $v$ is some increasing function. Risk aversion, $\mu$, is set to 2 for the baseline calibration. Estimates of the intertemporal labor supply elasticity, $\gamma$, typically range between 0 and 0.5 – see for example Altonji (1986) and Flood and MaCurdy (1992).[8] As the benchmark I set $\gamma = 0.3$ but I also consider a lower (0.1) and a higher (0.5) elasticity.

The effective potential labor supply depends on the size of the labor force (captured by $\eta$) and by its efficiency (captured by $\nu$). The fraction of individuals that is active in the labor market, $\eta$, is shown in Figure 1. People aged 20 to 64 are assumed to be workers.[9]

Worker efficiency is affected by the age structure of the labor force. Middle-aged workers appear to be both more productive (reflected by a higher wage rate) and to participate in the labor market to a higher extent than young and old workers. The variable $\nu$ captures these effects. Age-specific productivity is based on estimates for the United States reported in Hansen (1993). Participation rates are estimated by Fullerton (1999) and are also based on U.S. data. These age-specific values for productivity and participation (reported in Table I) are then multiplied by the number of workers in that age group relative to the total number of workers. Finally, $\nu_0$ is normalized to unity in all countries so that $\nu_t$ indicates the relative impact of the age-structure on efficiency from year 2000 to year $2000 + t$. Note that the same adjustment factor was used for all countries. In reality, age-specific participation rates may be quite different in different countries because of different education or retirement patterns. However, the quantitative importance of is small, so such differences are likely to be negligible.

**Table I**
**Calculation of $\nu$**

| Age | Productivity relative to average[a] | Participation rate[b] |
|---|---|---|
| 20-24 | 0.71 | 65.9[c] |
| 25-34 | 0.99 | 84.6 |
| 35-44 | 1.15 | 84.7 |
| 45-54 | 1.15 | 82.5 |
| 55-64 | 0.84 | 59.3 |

Notes: [a)] Hansen (1993) [b)] Fullerton (1999) [c)] The value refers to ages 16-24.

The consumption tax rate, $\tau^c$, and the initial tax rate on labor income, $\tau_0^h$, were taken from table 4 in Carey and Tchilinguirian (2000). They calculate effective average tax rates for OECD countries using an improved version of the method suggested by Mendoza et al. (1995). The first two columns in Table II summarize these country-specific tax rates.

[8] The intertemporal labor supply elasticity is equal to $\gamma$ when $\mu = 1$, and approximately equal to $\gamma$ otherwise. In practice, estimates of the elasticity are often estimates of $\gamma$ rather than the elasticity.

[9] The demographic forecasts are based on the United Nation's estimates from 1998. Thomas Lindh kindly provided this data.

**Table II**
**Country-specific parameters**

| | $\tau_0^h$ | $\tau^c$ | $\bar{d}$ | $\bar{b}$ | $\bar{g}$ | $w$ (%) | reduc. |
|---|---|---|---|---|---|---|---|
| Belgium | 0.397 | 0.187 | 1.110 | 0.163 | 0.192 | 1.53 | 0.087 |
| Denmark | 0.428 | 0.257 | 0.516 | 0.144 | 0.269 | 0.79 | 0.194 |
| Finland | 0.445 | 0.227 | 0.406 | 0.167 | 0.250 | 0.78 | 0.207 |
| France | 0.402 | 0.180 | 0.646 | 0.179 | 0.190 | 8.87 | 0.066 |
| Germany | 0.359 | 0.158 | 0.617 | 0.154 | 0.178 | 12.34 | 0.066 |
| Italy | 0.363 | 0.160 | 1.152 | 0.150 | 0.171 | 8.60 | 0.114 |
| Netherlands | 0.410 | 0.187 | 0.606 | 0.137 | 0.264 | 2.37 | 0.247 |
| Norway | 0.355 | 0.269 | 0.332 | 0.115 | 0.264 | 0.67 | 0.162 |
| Portugal | 0.227 | 0.205 | 0.554 | 0.108 | 0.170 | 1.48 | 0.114 |
| Spain | 0.304 | 0.137 | 0.706 | 0.129 | 0.156 | 5.95 | 0.114 |
| Sweden | 0.485 | 0.187 | 0.644 | 0.147 | 0.265 | 1.34 | 0.209 |
| U.K. | 0.210 | 0.169 | 0.492 | 0.147 | 0.112 | 8.83 | 0.023 |
| Canada | 0.287 | 0.131 | 0.825 | 0.070 | 0.192 | 4.68 | 0.117 |
| USA | 0.226 | 0.061 | 0.571 | 0.071 | 0.126 | 41.78 | 0.034 |
| World | 0.291 | 0.122 | 0.653 | 0.114 | 0.156 | 100.00 | 0.068 |

Note: $w$ is the country's population weight, reduc. is Adema's reduction factor.

Production is given by the Cobb-Douglas function

$$y = F(k, H) = k^{\theta} H^{1-\theta}.$$

### 3.1 Initial steady state

All economies are assumed to be in a steady state in year 2000. These steady states are calibrated to be similar to the actual economies in the recent past. I assume that the initial net position of households against the rest of the world is zero in each economy (see Nordin et. al., 1992, pp 30-32, for Swedish evidence), hence $a/y = d/y + k/y$. I further assume that $\tau^{\pi} = 0.4$ and $\tau^{a} = 0$ in all countries.[10]

The time discount rate, $\beta$, is calibrated so that the capital-output ratio equals 2.5 in all countries. The capital share in production, $\theta$, is set to .36, and the depreciation rate of capital, $\delta$, is set to 10 percent per year. Consequently $(1 + \tau^{\pi}) r = 0.044$.

Public transfers, $b$, are based on OECD's Social Expenditure Data Base. Transfers per capita for the initial steady state are calculated as the sum of public spending on old-age cash benefits, disability cash benefits, occupational injury and disease, sickness benefits, survivors pensions, family cash benefits, unemployment benefits, and housing benefits. The values are from 1995 or 1996 depending on availability, and all values are relative to GDP per capita. The transfers reported by OECD are gross and may be subject to taxation in some countries. The adjustment

[10] Estimates of tax rates on capital income varies substaintially between studies and appear unreliable. However, estimates around 40 percent are common, see for example Carey and Tchilinguirian (2000). If $\tau^{a} > 0$ the government will face higher interest rates than households. Since $\tau^{a}$ is not a policy variable, an optimizing government facing exogenous interest rates will then accumulate more and more assets as a substitute for household savings. To avoid such implications, I set $\tau^{a} = 0$.

factors reported in Adema's (1999) table 3, row 1, have therefore been used to adjust the OECD figures.[11]

Public debt is gross government debt in year 2000 from OECD's Economic Outlook, relative to GDP from the same data set. Table II reports the country-specific parameter values used in the initial steady-state.[12]

*The world economy*

It is assumed that labor supply is 33 percent of available time in the initial steady state for the world economy, and that $\beta R = 1$ (otherwise no steady state would exist under optimal policy). For any variable $x$, let $\bar{x} \equiv x/y$, and let $C \equiv \eta c^a + (1-\eta)\, c^i$. The seven equations below then determine the remaining variables in the steady state, $c^a$, $c^i$, $\zeta$, $w$, $y$, $g$, and $r$,

$$(1+\tau^c)\,\bar{C} = (1-\tau^a)\, r\bar{a} + \left(1-\tau^h\right)\eta\nu h\bar{w} + \bar{b} \tag{13}$$

$$\bar{g} + \bar{b} + r\bar{d} = \tau^h \eta\nu h\bar{w} + \tau^a r\bar{a} + \tau^\pi r\bar{k} + \tau^c \bar{C} \tag{14}$$

$$\bar{k} = \frac{\theta}{(1+\tau^\pi)\, r + \delta} \tag{15}$$

$$y = \bar{k}^{\frac{\theta}{1-\theta}} \eta\nu h \tag{16}$$

$$\bar{w} = \frac{1-\theta}{\eta\nu h} \tag{17}$$

$$\frac{U_1}{\eta} = \frac{U_2}{1-\eta} \tag{18}$$

$$\frac{\left(1-\tau^h\right)\bar{w}\nu y}{1+\tau^c} = -\frac{U_3}{U_1} \tag{19}$$

The seven equations above are the household budget constraint; the government budget constraint; the production function; the first order conditions for factor prices (two equations); and the first order conditions for $c^i$ and $h$.

Table III summarizes the parameter values that are common to all economies.

**Table III**
**Parameter values and initial steady state**

| | | |
|---|---|---|
| Risk aversion | $\mu$ | 2.000 |
| Labor-supply elasticity | $\gamma$ | 0.300 |
| Time discount rate | $\beta$ | 0.969 |
| Capital-output ratio | $\bar{k}$ | 2.500 |
| Capital share | $\theta$ | 0.360 |
| Interest rate | $r$ | 0.031 |
| Hours worked, percent of available time | $h$ | 0.330 |
| Tax on firm profits | $\tau^\pi$ | 0.400 |
| Tax on return to capital | $\tau^a$ | 0.000 |

[11] Adema does not report adjustment factors for France, Spain and Portugal. The German adjustment factor was used for France, while the Italian factor was used for Spain and Portugal.

[12] The levels of public consumption reported in the table are solved from the equilibrium conditions as described below.

Initial public debt and public policy in the world economy are assumed to be the population weighted average of debt and policy in the U.S., Canada, and the twelve examined European countries. The population weights and resulting world averages are also reported in Table II.

*Small open economies*

Equations (13) - (19) are also used to find the initial steady states for small economies. In these economies, however, the preference parameter $\zeta$ is set to the same value as in the world economy, and $h$ is not calibrated.

### 3.2 Future development of public expenditure

Two alternative approaches are used for choosing the future paths of transfers and public consumption. Both approaches attempt to predict what happens if there are no changes in policies. The first approach assumes that these variables remain constant at the level from the initial steady state. The implicit assumptions are then that costs grow proportionally with technological development (recall that there is no such development in the model), and that costs are independent of the age structure in the population.

The alternative approach tries to predict how the demographic development will affect public expenditure. Per capita levels of public expenditure in the initial steady state are then calculated separately for the young, for workers, and for the old. These per capita levels are then assumed to remain constant, but total expenditure will be affected by the demographic composition of the population. When calculating the group-specific transfer levels, it is assumed that disability cash benefits, survivors pensions, and housing benefits are evenly divided in the full population, that family cash benefits are evenly divided among the young and the workers, that old-age cash benefits only accrues to the old, and that occupational injury and disease benefits, sickness benefits, and unemployment benefits only accrues to the workers. Many items in public consumption are (implicitly) assumed to be independent of the age structure (for example police and defence). Health care costs, education costs, and spending on day care and long term care are allocated to the different age groups.

Health care costs are assumed to be four times higher per capita for old persons (aged 65+) than for others (see Batljan and Lagergren, 2000, section 3.2) . Further, it is assumed that only the old consume long term care, only the young consume day care and primary and secondary education, and only workers consume tertiary education. The data sources are OECD (1998, table VI.3.) for health care and long term care, and OECD (2001, table B2) for education costs.[13]

The resulting age structure of transfers and public consumption is reported in Tables IV and V. The age structure of public consumption in Sweden is similar to that reported in Olsson and Nordén (1999). The calculations for transfers, however, appear to assign considerably smaller values to children than what Olsson and Nordén did.

[13] Public spending on long term care is not reported for all countries. It is assumed to be 50 percent of total spending in Italy, Spain, and Portugal, and 80 percent in Denmark.

**Table IV**
**Age structure of transfers**

| | per capita frac. of gdp per capita | | | fraction of total transfers year 2000 | | | fraction of total transfers year 2030 | | |
|---|---|---|---|---|---|---|---|---|---|
| | young | worker | old | young | worker | old | young | worker | old |
| Belgium | 0.063 | 0.119 | 0.455 | 0.090 | 0.443 | 0.467 | 0.067 | 0.336 | 0.597 |
| Denmark | 0.041 | 0.112 | 0.431 | 0.065 | 0.481 | 0.454 | 0.052 | 0.363 | 0.585 |
| Finland | 0.069 | 0.132 | 0.469 | 0.101 | 0.479 | 0.419 | 0.075 | 0.341 | 0.584 |
| France | 0.061 | 0.104 | 0.646 | 0.086 | 0.340 | 0.574 | 0.063 | 0.256 | 0.681 |
| Germany | 0.030 | 0.078 | 0.605 | 0.041 | 0.316 | 0.642 | 0.026 | 0.210 | 0.764 |
| Italy | 0.040 | 0.061 | 0.572 | 0.051 | 0.255 | 0.694 | 0.030 | 0.163 | 0.806 |
| Netherlands | 0.049 | 0.111 | 0.406 | 0.086 | 0.505 | 0.409 | 0.056 | 0.348 | 0.596 |
| Norway | 0.050 | 0.083 | 0.342 | 0.112 | 0.427 | 0.461 | 0.086 | 0.333 | 0.580 |
| Portugal | 0.036 | 0.065 | 0.383 | 0.076 | 0.367 | 0.558 | 0.049 | 0.288 | 0.663 |
| Spain | 0.024 | 0.076 | 0.453 | 0.039 | 0.364 | 0.596 | 0.022 | 0.263 | 0.715 |
| Sweden | 0.055 | 0.107 | 0.407 | 0.089 | 0.429 | 0.482 | 0.067 | 0.333 | 0.600 |
| U.K. | 0.074 | 0.095 | 0.448 | 0.126 | 0.384 | 0.490 | 0.096 | 0.303 | 0.601 |
| Canada | 0.017 | 0.043 | 0.306 | 0.062 | 0.377 | 0.560 | 0.040 | 0.243 | 0.717 |
| USA | 0.012 | 0.023 | 0.425 | 0.048 | 0.196 | 0.755 | 0.028 | 0.125 | 0.847 |
| World | 0.031 | 0.058 | 0.477 | 0.061 | 0.287 | 0.652 | 0.040 | 0.199 | 0.761 |

**Table V**
**Age structure of public consumption**

| | per capita frac. of gdp per capita | | | fraction of total transfers year 2000 | | | fraction of total transfers year 2030 | | |
|---|---|---|---|---|---|---|---|---|---|
| | young | worker | old | young | worker | old | young | worker | old |
| Belgium | 0.273 | 0.132 | 0.297 | 0.328 | 0.414 | 0.258 | 0.284 | 0.375 | 0.341 |
| Denmark | 0.419 | 0.184 | 0.387 | 0.360 | 0.421 | 0.218 | 0.329 | 0.387 | 0.283 |
| Finland | 0.354 | 0.186 | 0.337 | 0.347 | 0.452 | 0.201 | 0.309 | 0.412 | 0.279 |
| France | 0.271 | 0.125 | 0.302 | 0.359 | 0.387 | 0.253 | 0.315 | 0.353 | 0.332 |
| Germany | 0.253 | 0.117 | 0.313 | 0.300 | 0.412 | 0.288 | 0.245 | 0.355 | 0.400 |
| Italy | 0.283 | 0.119 | 0.229 | 0.321 | 0.435 | 0.244 | 0.266 | 0.402 | 0.332 |
| Netherlands | 0.281 | 0.173 | 0.428 | 0.287 | 0.461 | 0.252 | 0.231 | 0.396 | 0.373 |
| Norway | 0.356 | 0.172 | 0.465 | 0.344 | 0.384 | 0.272 | 0.301 | 0.349 | 0.351 |
| Portugal | 0.284 | 0.116 | 0.215 | 0.382 | 0.420 | 0.198 | 0.322 | 0.415 | 0.263 |
| Spain | 0.250 | 0.104 | 0.232 | 0.333 | 0.414 | 0.253 | 0.256 | 0.390 | 0.353 |
| Sweden | 0.389 | 0.170 | 0.416 | 0.349 | 0.377 | 0.273 | 0.303 | 0.347 | 0.350 |
| U.K. | 0.189 | 0.051 | 0.217 | 0.422 | 0.267 | 0.311 | 0.350 | 0.231 | 0.419 |
| Canada | 0.257 | 0.137 | 0.325 | 0.343 | 0.441 | 0.216 | 0.290 | 0.385 | 0.325 |
| USA | 0.176 | 0.075 | 0.254 | 0.397 | 0.351 | 0.252 | 0.312 | 0.308 | 0.380 |
| World | 0.225 | 0.098 | 0.272 | 0.365 | 0.376 | 0.258 | 0.297 | 0.334 | 0.369 |

Obviously, allocating the different components of public expenditure to specific age groups is problematic and associated with many approximations and assumptions. Moreover, even if today's expenditure could be allocated to different age groups with precision, the age structure

of spending may change over time. Health care and long term care are important components in public expenditure – the amount to a significant fraction of public expenditure and the costs are mostly attributed to the old. When life expectancy increases, it is quite possible that the young retirees will become healthier so that the per capita consumption of these services will fall for the average retiree. The calculations in the alternative approach will then overestimate the demographic impact on public expenditure. Arguably, the two approaches for calibrating the path of future public expenditure result in one too optimistic path (the first approach) and one too pessimistic path (the second approach).

## 4 Findings

Figure 5 and the top left panel in Figure 6 show the interest rate path and the sequence of taxes that balances the public budget in the 'world economy' (the solution to equation [12]) under the benchmark specification. As expected, interest rates fall during the transition to a new steady state with an older population. In the long run, interest rates return to the equilibrium level. Figure 6 also reports implications on public and private asset holdings, total output, savings ratios, and labor supply. Figure 6 shows the interest path that solves this system of equations.

Figures 7A-7N show the optimal paths for debt and taxes in the respective countries, and resulting effects on these economies.[14] The top right panel in the figure shows that the optimal policy is to reduce debt during the initial years, before the demographic situation deteriorates. The dashed lines in Figure 7 show the implications of optimal policy when world market prices are assumed to be constant. Note first that the implied optimal policy is almost identical to the optimal policy in the benchmark scenario when the interest rate responds to the demographic change. The model was also solved with different values of the risk aversion and labor-supply elasticity, but the optimal policy is consistently similar to the one shown in Figure 7. With the alternative assumption for the development of public expenditures, the implied policy is an even sharper reduction in debt, accompanied by a higher initial tax raise.

Note also that although the private savings would be higher if the interest rate did not respond to the population aging, this effect on savings is not reflected in production levels. Except for the very first years, output is higher in the benchmark economy. Two factors cause this result. First, interest rates are low during the population aging episode because the number of workers is falling, implying a higher capital-output ratio and thus making workers more productive. Second, lower interest rates reduce interest payments on public debt so that less taxation is required in the benchmark economy.

The optimal policy in terms of tax rates is to increase taxes in the first year and then letting them remain approximately unchanged. This result is not surprising and is in line with the tax smoothing argument put forward in Barro (1979).

Tables VI to VIII report the implications of optimal policy for the respective countries. Results are reported for the benchmark specification of the economy, for the alternative assumption about the path of public expenditure, and for the assumption of constant world market prices. We see that the implied optimal policies are similar for most countries. Table VI shows that the optimal, immediate, tax increase varies from 2.1 percent for the U.K. to 8.3 percent for Finland. Table VII shows that average annual budget surpluses should be between 1.3 percent for the U.K. and 4.1 percent for the Netherlands during the first ten years and similar during the following decades. Finally, Table VIII shows the implied reductions in debt levels.

[14] Results for Canada and the United States are also reported here. Note, however, that these countries are not considered part of the study, and the small-economy assumptions are not plausible for the United States.

The differences between the extreme countries – the U.K. which appears least affected by the demographic change and Finland, Italy, and the Netherlands, which appear most affected – are explained by several factors. First, the size of the public sector matters for the effectiveness and distortions of taxation. The excess burden of taxation increases with the size of the government sector, thus taxes are more distortive in countries with a large government sector, such as Finland, and the effect on tax revenue of a one percent increase in tax rates is smaller. Second, population aging has a larger impact on the public budget in countries with high public expenditure. Third, in countries with a large initial public debt, such as Italy, the optimal policy is to reduce this debt significantly (see Table VIII) in order to lessen the future fiscal burden. Fourth, Figure 1 showed that the demographic change will be somewhat less severe in the U.K. than in many other countries. Finally, some countries (most notably the Netherlands and Belgium) are predicted to have a particularly unfavorable development of the age-structure within the group of workers. Table IX shows how changes in the age-structure of the work force is predicted to affect long-run productivity in different countries.

**Table VI**
**Implications of optimal policy: initial tax effect**

| | benchm. | $\{g_t, b_t\}$ | const. $r$ | $\gamma = 0.1$ | $\gamma = 0.5$ | $\mu = 1$ |
|---|---|---|---|---|---|---|
| Belgium | 5.4 | 11.7 | 6.9 | 5.3 | 5.5 | 6.2 |
| Denmark | 6.9 | 14.0 | 8.4 | 6.8 | 7.1 | 7.8 |
| Finland | 8.3 | 17.2 | 9.7 | 7.8 | 8.9 | 9.1 |
| France | 4.6 | 13.0 | 5.9 | 4.5 | 4.7 | 5.4 |
| Germany | 4.4 | 16.2 | 5.4 | 4.6 | 4.2 | 5.0 |
| Italy | 6.8 | 20.6 | 8.0 | 7.0 | 6.7 | 7.4 |
| Netherlands | 8.1 | 20.5 | 9.2 | 7.9 | 8.4 | 8.7 |
| Norway | 5.2 | 11.0 | 6.3 | 5.3 | 5.2 | 5.8 |
| Portugal | 2.3 | 7.2 | 3.0 | 2.7 | 2.0 | 2.7 |
| Spain | 4.1 | 13.6 | 4.8 | 4.5 | 3.8 | 4.5 |
| Sweden | 5.4 | 12.4 | 7.0 | 5.2 | 5.7 | 6.4 |
| U.K. | 2.1 | 6.3 | 2.9 | 2.3 | 2.0 | 2.6 |
| Canada | 3.7 | 8.9 | 4.8 | 3.9 | 3.6 | 4.3 |
| USA | 1.2 | 5.9 | 2.1 | 1.3 | 1.2 | 1.7 |

The table shows the tax increase, in percentage points, the first year with optimal policy. $\{g_t, b_t\}$ is the specification where public expenditure depends on the population's age-structure.

**Table VII**
**Implications of optimal policy: necessary budget surplus**

| | until 2010 | | until 2030 | |
|---|---|---|---|---|
| | benchmark | $\{g_t, b_t\}$ | benchmark | $\{g_t, b_t\}$ |
| Belgium | 3.1 | 6.6 | 3.0 | 6.1 |
| Denmark | 2.8 | 6.0 | 2.1 | 4.5 |
| Finland | 3.4 | 7.2 | 2.3 | 4.7 |
| France | 2.5 | 7.0 | 2.3 | 6.0 |
| Germany | 2.7 | 7.8 | 2.7 | 7.2 |
| Italy | 4.0 | 10.7 | 4.2 | 10.6 |
| Netherlands | 4.1 | 10.1 | 3.7 | 8.4 |
| Norway | 2.5 | 5.9 | 2.1 | 4.8 |
| Portugal | 2.0 | 5.2 | 2.6 | 6.1 |
| Spain | 3.4 | 9.2 | 4.2 | 10.2 |
| Sweden | 2.5 | 5.7 | 2.7 | 5.0 |
| U.K. | 1.3 | 4.1 | 1.4 | 3.7 |
| Canada | 2.3 | 5.5 | 2.0 | 4.5 |
| USA | 0.9 | 4.0 | 0.7 | 3.2 |

The table shows the average annual budget surplus (in percentage points) implied by optimal policy.

**Table VIII**
**Implications of optimal policy: change in public debt**

| | until 2010 | | until 2030 | |
|---|---|---|---|---|
| | benchmark | $\{g_t, b_t\}$ | benchmark | $\{g_t, b_t\}$ |
| Belgium | −30.0 | −60.9 | −86.8 | −182.7 |
| Denmark | −26.5 | −56.6 | −65.2 | −135.7 |
| Finland | −32.8 | −69.2 | −71.3 | −144.8 |
| France | −24.1 | −65.1 | −68.0 | −179.1 |
| Germany | −26.2 | −73.4 | −81.4 | −222.9 |
| Italy | −38.7 | −97.2 | −124.6 | −324.8 |
| Netherlands | −40.0 | −94.7 | −116.1 | −264.6 |
| Norway | −24.0 | −56.8 | −64.1 | −147.6 |
| Portugal | −20.5 | −50.6 | −81.0 | −184.6 |
| Spain | −35.5 | −88.2 | −131.1 | −319.3 |
| Sweden | −23.7 | −52.6 | −80.1 | −150.9 |
| U.K. | −13.1 | −39.6 | −40.2 | −120.9 |
| Canada | −22.5 | −52.2 | −57.0 | −135.1 |
| USA | −8.4 | −38.4 | −20.8 | −95.0 |

The table shows the increase in debt (in percentage points) implied by optimal policy.

**Table IX**
**Impact of age-composition of work force on long-run productivity**

| | $\Delta\nu_{\text{long run}}$ |
|---|---|
| Belgium | −4.3 |
| Denmark | −2.7 |
| Finland | −3.6 |
| France | −3.3 |
| Germany | −3.0 |
| Italy | −3.4 |
| Netherlands | −5.4 |
| Norway | −3.7 |
| Portugal | −1.7 |
| Spain | −2.2 |
| Sweden | −4.1 |
| U.K. | −3.4 |
| Canada | −4.4 |
| USA | −4.1 |

The table shows the percentage change in $\nu$.

### 4.1 Alternative policies and welfare effects

How important is it that the government tries to follow the optimal debt strategy? Would welfare be significantly reduced if mistakes were made or if the government pursued other objectives? To answer these questions, the optimal policy was compared to a policy balancing the public budget in each period.[15] The alternative policy is thus a sequence of tax rates, $\left\{\hat{\tau}_t^h\right\}$, that fulfils equation (10) for each $t$ under the assumption that the sequences for interest rates and wages are exogenous.

The dotted lines in the top-left panel of Figures 7 show the sequences of tax rates that balances the each country's budget. While tax rates can be held down initially, substantial increases are required between years 2020 and 2050 when the number of retirees increases. Consequently, hours worked and output is lower in the long run with the balanced-budget policy. Again, the corresponding figures for all other countries are similar.

Although the balanced-budget policy is quiet different from the optimal policy, the difference in households' welfare turns out to be modest. Table X shows welfare losses for different specifications of the model. As expected, tax smoothing is more important when labor supply is elastic. Still, a welfare loss of 0.5 percent of annual consumption amounts to less than USD 100 per person and year. With labor-supply elasticities like those typically used by macroeconomists, the welfare losses of deviations from the optimal policy would be sizeable.

[15] Note that the study ignores business cycle fluctuations. A balanced-budget policy in the model economy is therefore less drastic than a real-world ditto.

**Table X**
**Welfare loss with balanced-budget policy**

| | benchmark | $\{g_t, b_t\}$ | constant $r$ | $\gamma = 0.1$ | $\gamma = 0.5$ | $\mu = 1$ |
|---|---|---|---|---|---|---|
| Belgium | 0.24 | n.s. | 0.29 | 0.06 | 0.75 | 0.31 |
| Denmark | 0.21 | n.s. | 0.29 | 0.05 | 0.67 | 0.30 |
| Finland | 0.36 | n.s. | 0.51 | 0.07 | n.s. | 0.52 |
| France | 0.13 | n.s. | 0.16 | 0.03 | 0.31 | 0.17 |
| Germany | 0.12 | n.s. | 0.15 | 0.03 | 0.28 | 0.16 |
| Italy | 0.61 | n.s. | 0.52 | 0.11 | n.s. | 0.54 |
| Netherlands | 0.44 | n.s. | 0.77 | 0.10 | n.s. | 0.80 |
| Norway | 0.09 | 0.78 | 0.10 | 0.03 | 0.19 | 0.11 |
| Portugal | 0.06 | 0.43 | 0.05 | 0.02 | 0.10 | 0.06 |
| Spain | 0.25 | n.s. | 0.26 | 0.07 | 0.56 | 0.28 |
| Sweden | 0.51 | n.s. | 0.69 | 0.08 | n.s. | 0.73 |
| U.K. | 0.01 | 0.10 | 0.01 | 0.00 | 0.02 | 0.01 |
| Canada | 0.05 | 0.30 | 0.06 | 0.02 | 0.08 | 0.06 |
| USA | 0.00 | 0.08 | 0.01 | 0.00 | 0.01 | 0.01 |

Note: Welfare loss in percent of annual consumption. N.s. = no solution with balanced-budget policy.

The results reported in Table X also indicate that under the alternative assumption of age-dependent public expenditure, pursuing a balanced-budget policy will be infeasible in most countries. This is due to a Laffer-curve effect. To balance the budget, year-to-year fluctuations in public expenditure and in the tax base, may require sharp fluctuations in the tax rate. But if taxes are already high, further tax increase may induce households to substitute labor supply into periods with lower taxes. Thus only the countries with low public expenditure (the U.K. and Portugal) or a small initial debt (Norway) appear able to balance the budget in that scenario.

Cutler et al. (1990) argue that although the optimal policy for the U.S. government probably is to reduce the public debt in the years before the dependency ratio deteriorates, the welfare gains of such a policy are likely to be small since taxes are not particularly distortionary. The present study supports their story, but also indicates that their arguments are not valid for the typical European countries, where the public sector is larger and where the demographic development is somewhat more problematic.

## 5 Concluding remarks

The above analysis ignores several factors that can have important effects on public finances in the future. For example, the process of increased internationalization, tax competition between EU countries, and a more mobile labor force can make the collection of taxes more difficult. Such changes would be similar to an increased labor-supply elasticity, making taxes more distortive over time. Taking these factors into account would therefore make the case for reducing the debt today even stronger.

Implicitly, the study has also assumed that the generosity and structure of welfare and pension systems and public services are unaffected by the demographic change. The internationalization and population aging may imply that the generosity of welfare systems must be reduced or that welfare systems must be reformed. Households may then respond by increasing

savings and labor supply, and thus reduce the importance of debt reduction today.

Furthermore, maintaining an unaltered level of public services may be difficult when the demographic structure changes. For example, even if consumption needs for children and old are identical and do not change over time, population aging requires investments in the infrastructure (schools cannot easily be transformed into hospitals). Evidence reported in Bucht et al. (2000) shows that, at least in the short run, school expenditure does not seem to vary directly with the number of students.

There is little doubt that most developed countries will see an aging of their populations in the coming 30 to 40 years. Still, forecasting the long-term future is difficult and associated with many sources of uncertainty. Lindh (2001) reports that previous forecasts of survival probabilities have been biased downwards, and that nativity is volatile but difficult to predict. Nativity may also be influenced by policy, something which has been neglected here. Migration is another factor that is difficult to forecast (for example, migration flows may be influenced by war episodes or changes in immigration policies) but that can influence demographic structures.

Considering the above caveats, the implications of the model and experiments conducted should be interpreted cautiously. The following conclusions, however, do not appear daring. First, population aging will put pressure on future public finances. In the long run taxes must be raised or public spending cut. Second, at least if intergenerational issues are ignored, welfare is enhanced if public debts are reduced during the next two or three decades. Finally, the welfare benefits of such policies compared to policies that hold debts constant are likely to be small or modest, except in countries with a large public sector and/or a large public debt.

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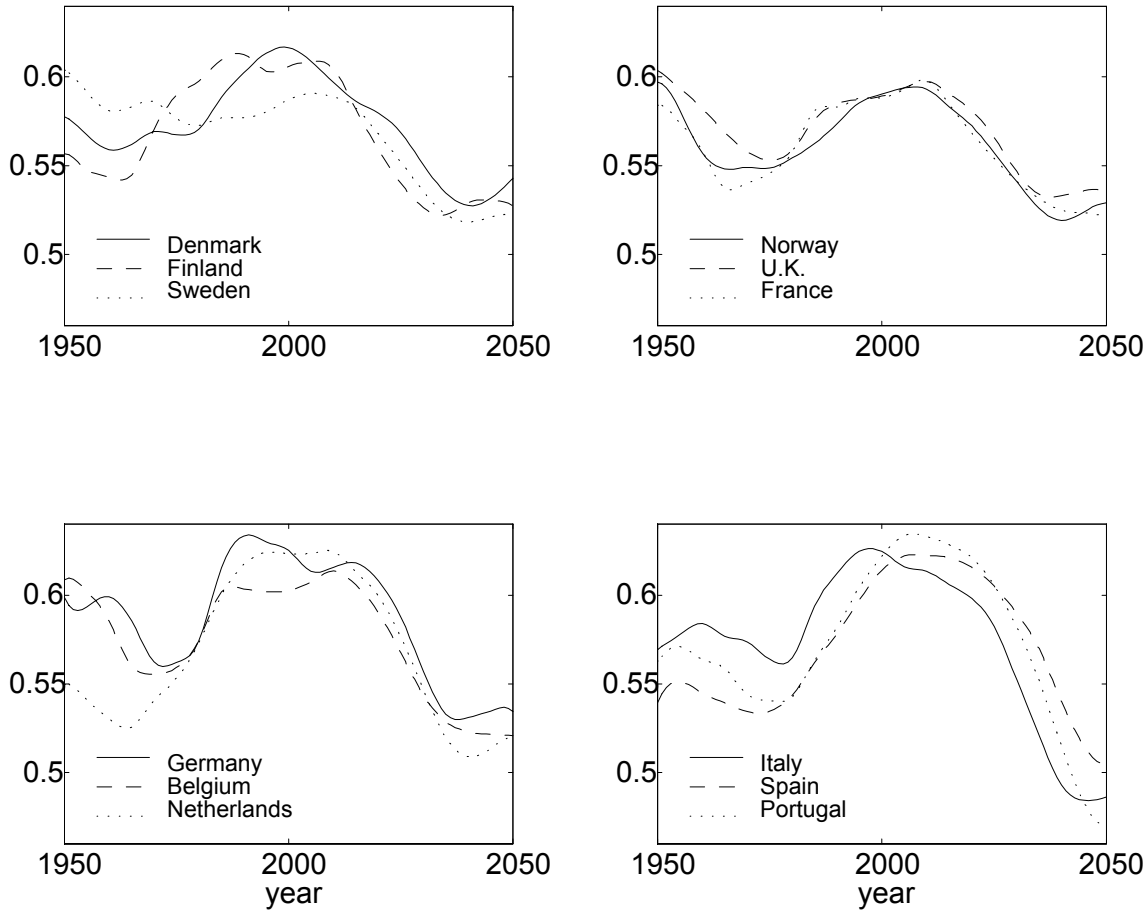


Figure 1: Share of people in working age (20-64)

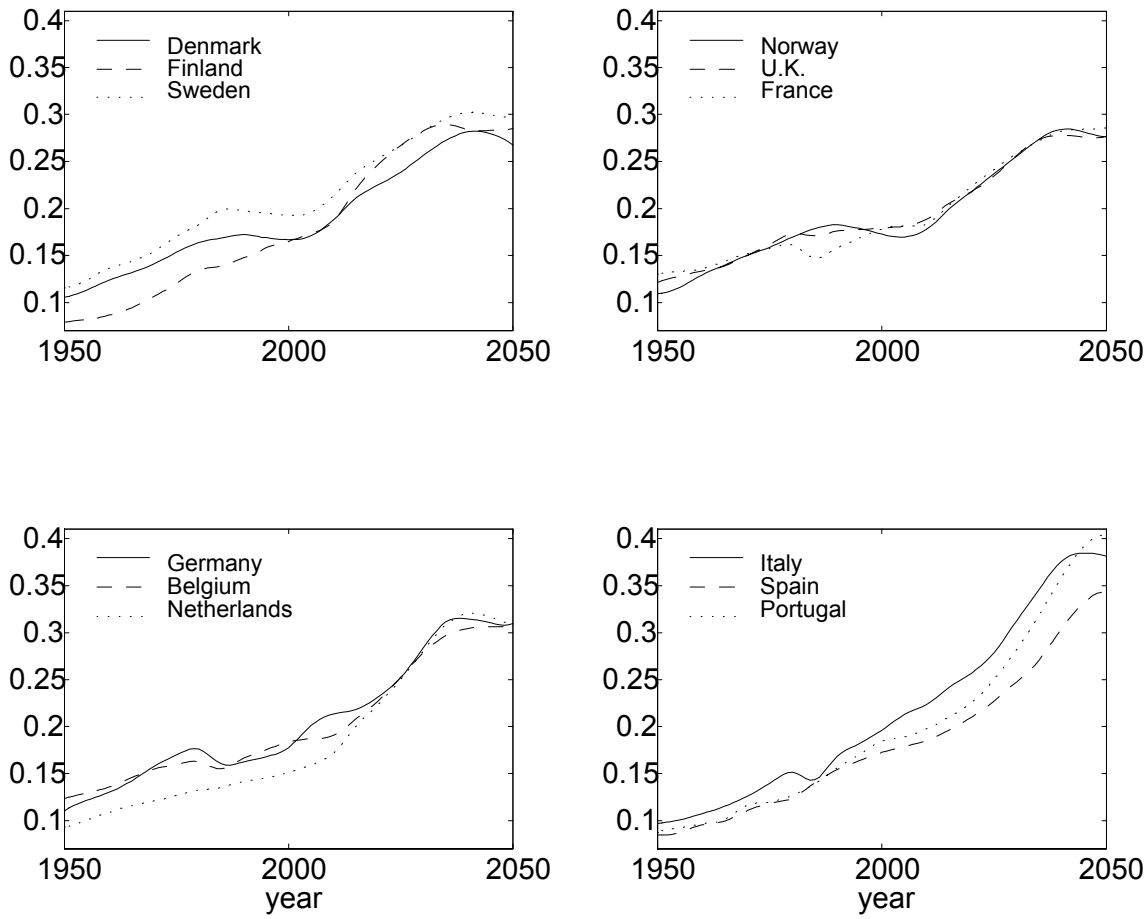


Figure 2: Share of old (aged 65+) in population

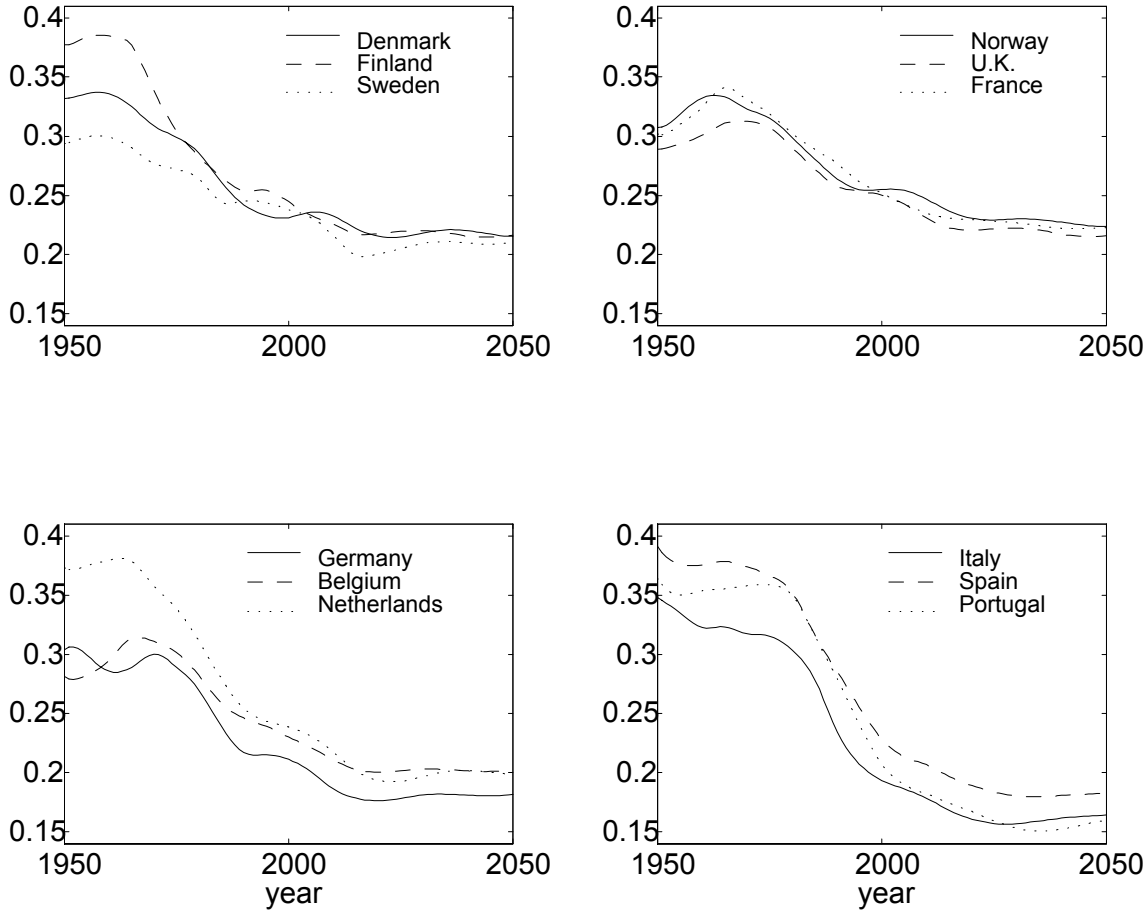


Figure 3: Share of young (aged 0-19) in population

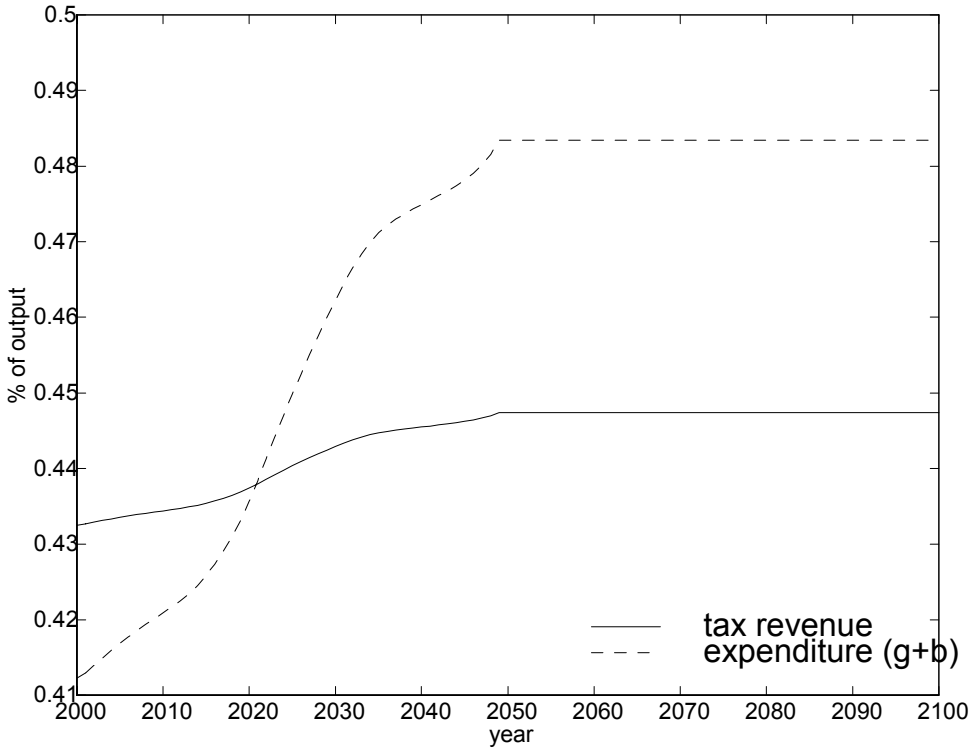


Figure 4: Direct impact of demographic change on public budget (Sweden)

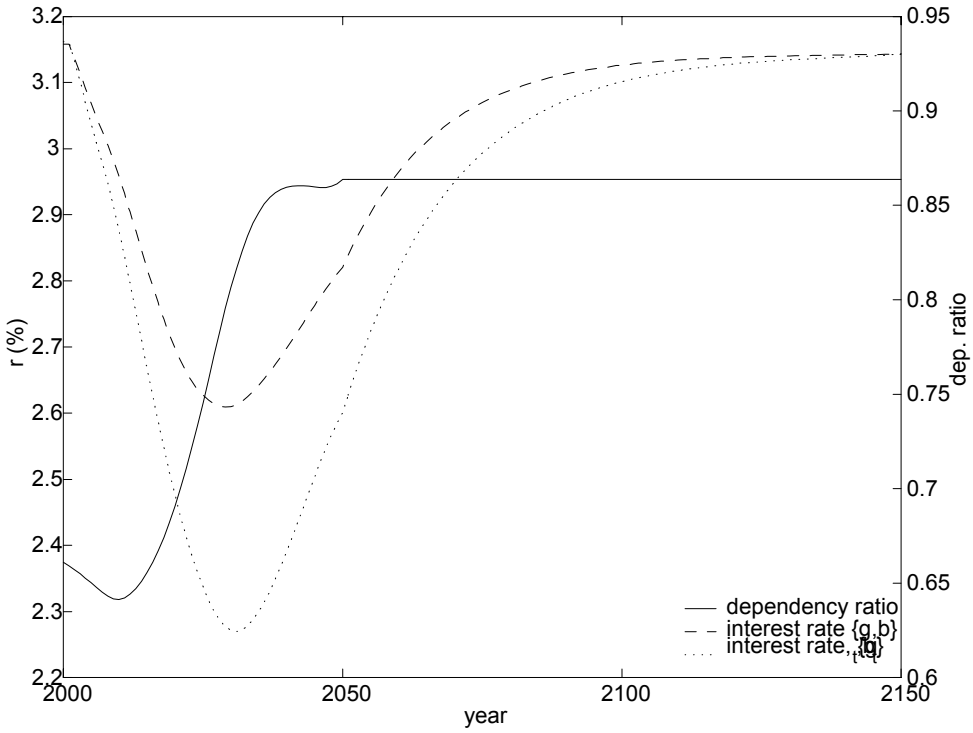


Figure 5: Interest rate

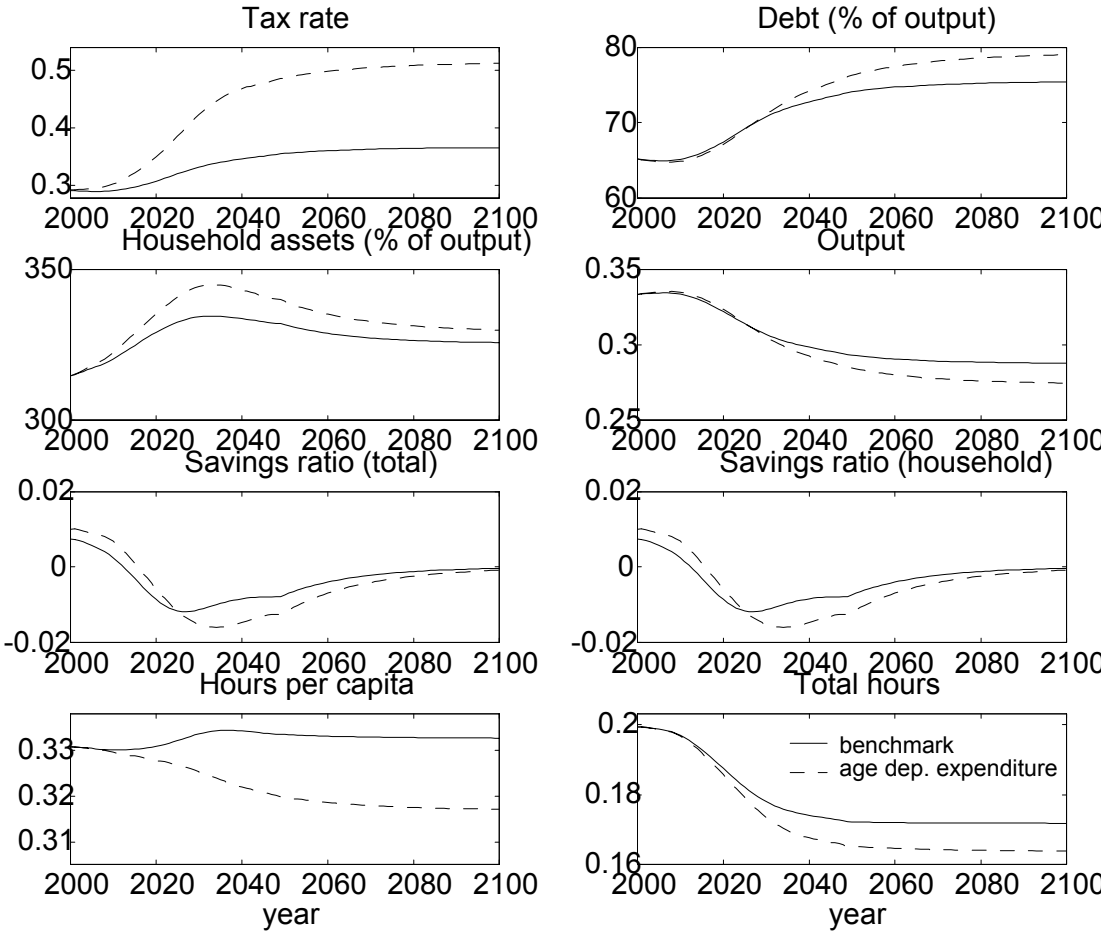


Figure 6: World economy with balance-budget policy

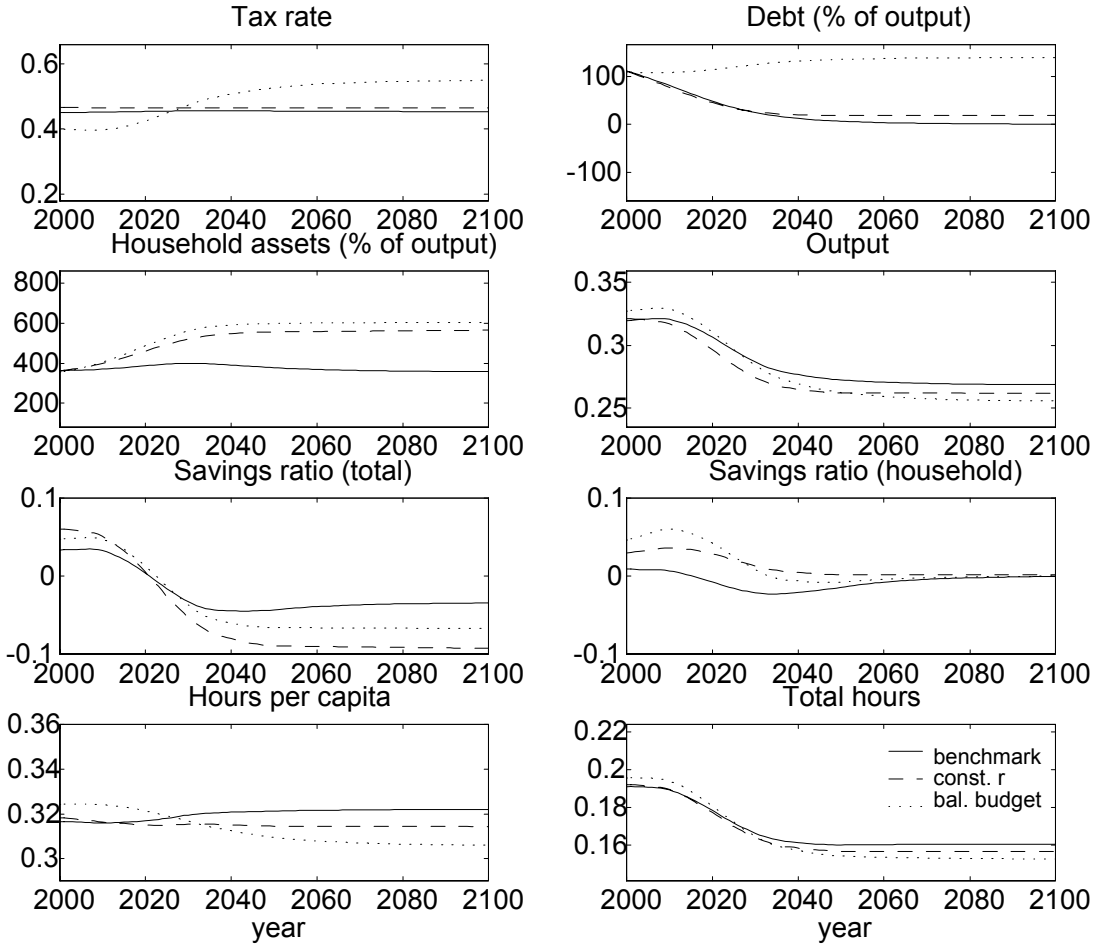


Figure 7A: Optimal policy vs. balanced-budget policy, Belgium

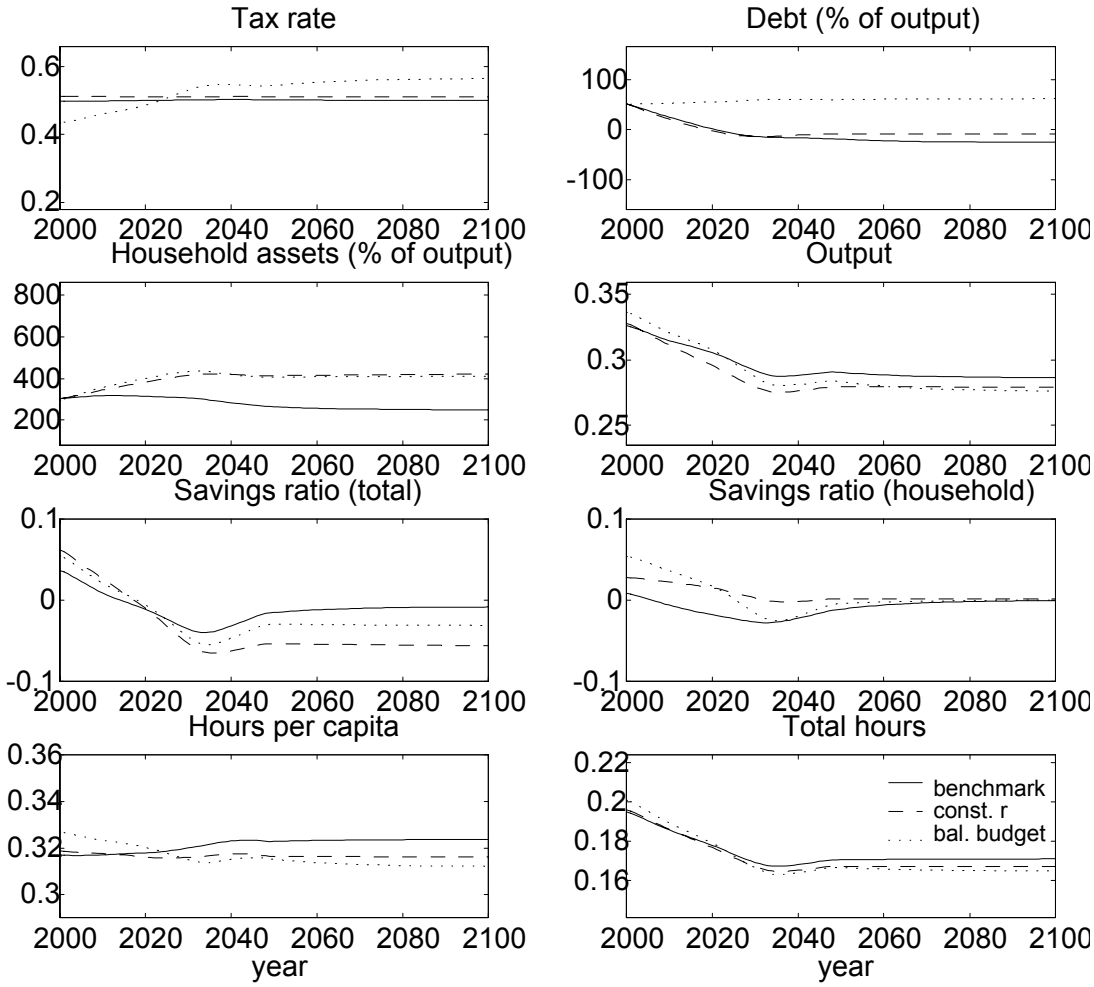


Figure 7B: Optimal policy vs. balanced-budget policy, Denmark

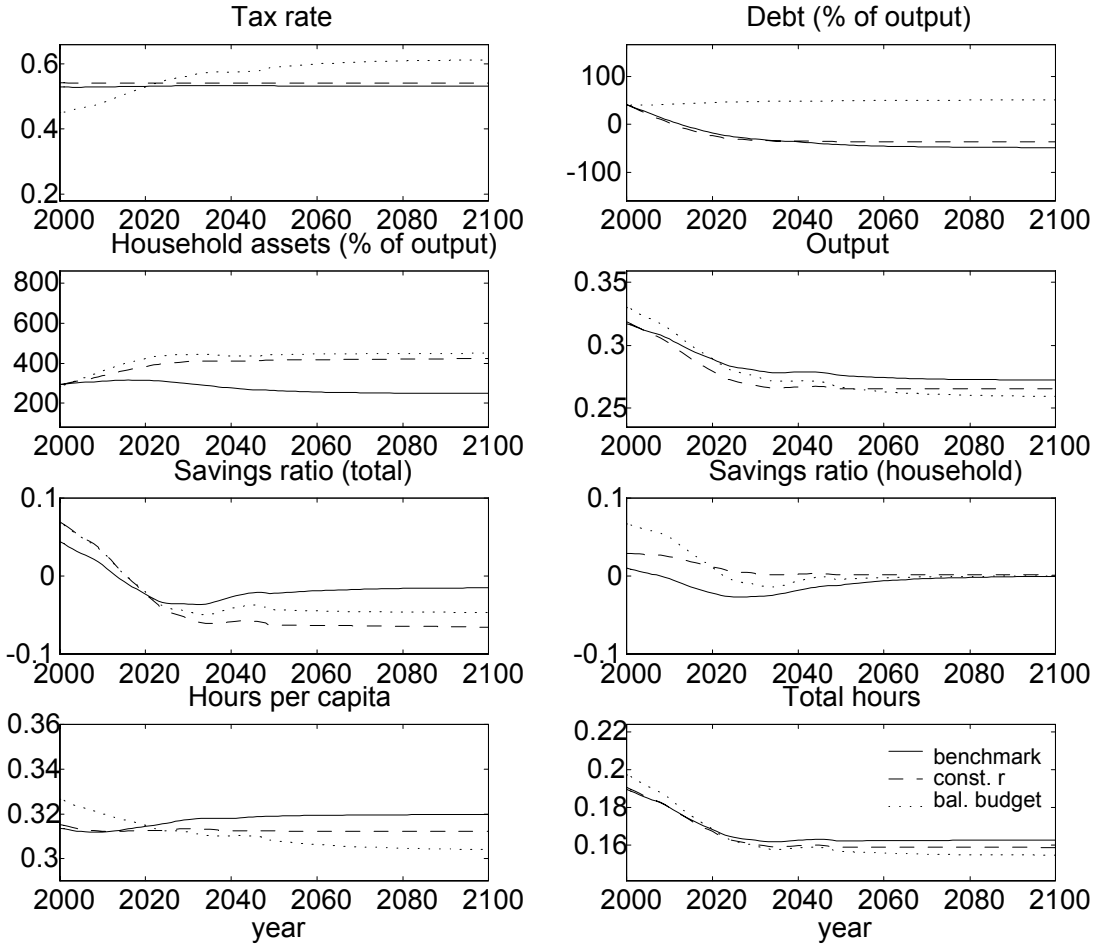


Figure 7C: Optimal policy vs. balanced-budget policy, Finland

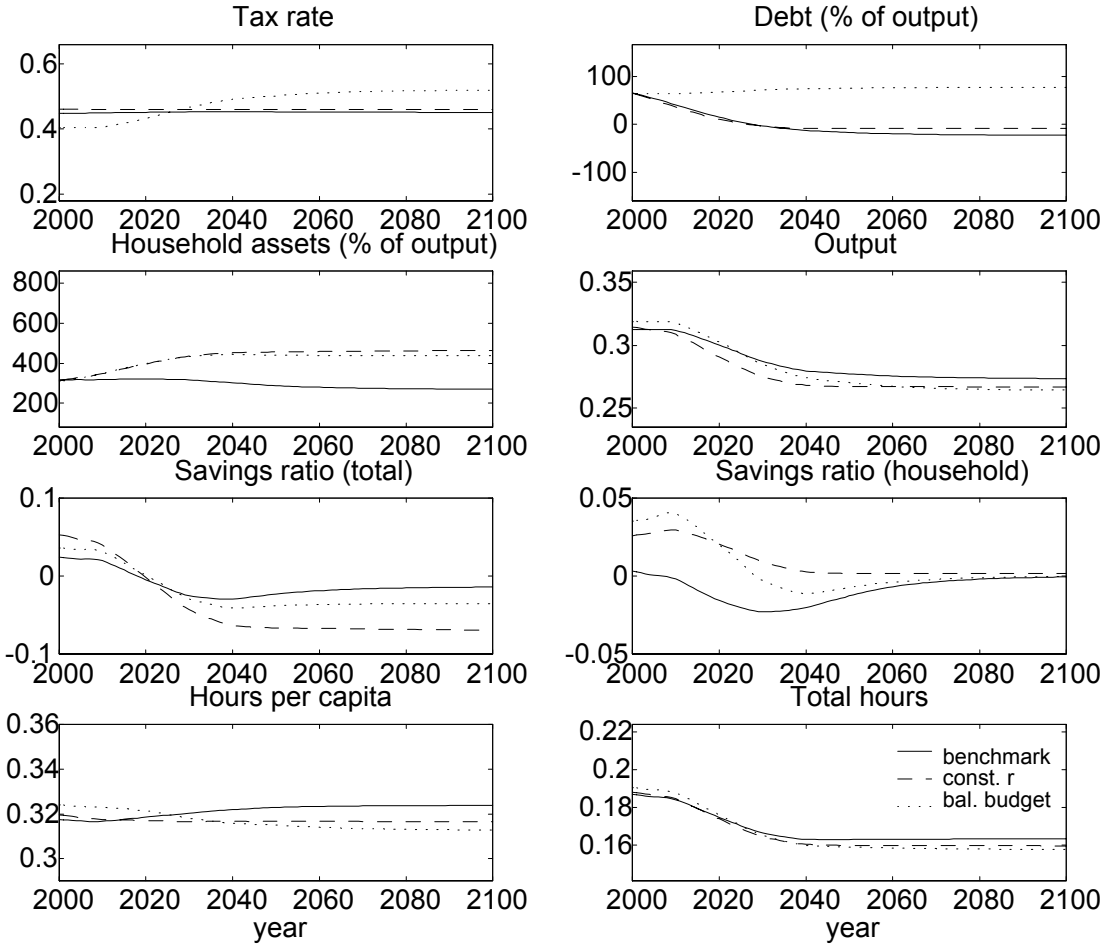


Figure 7D: Optimal policy vs. balanced-budget policy, France

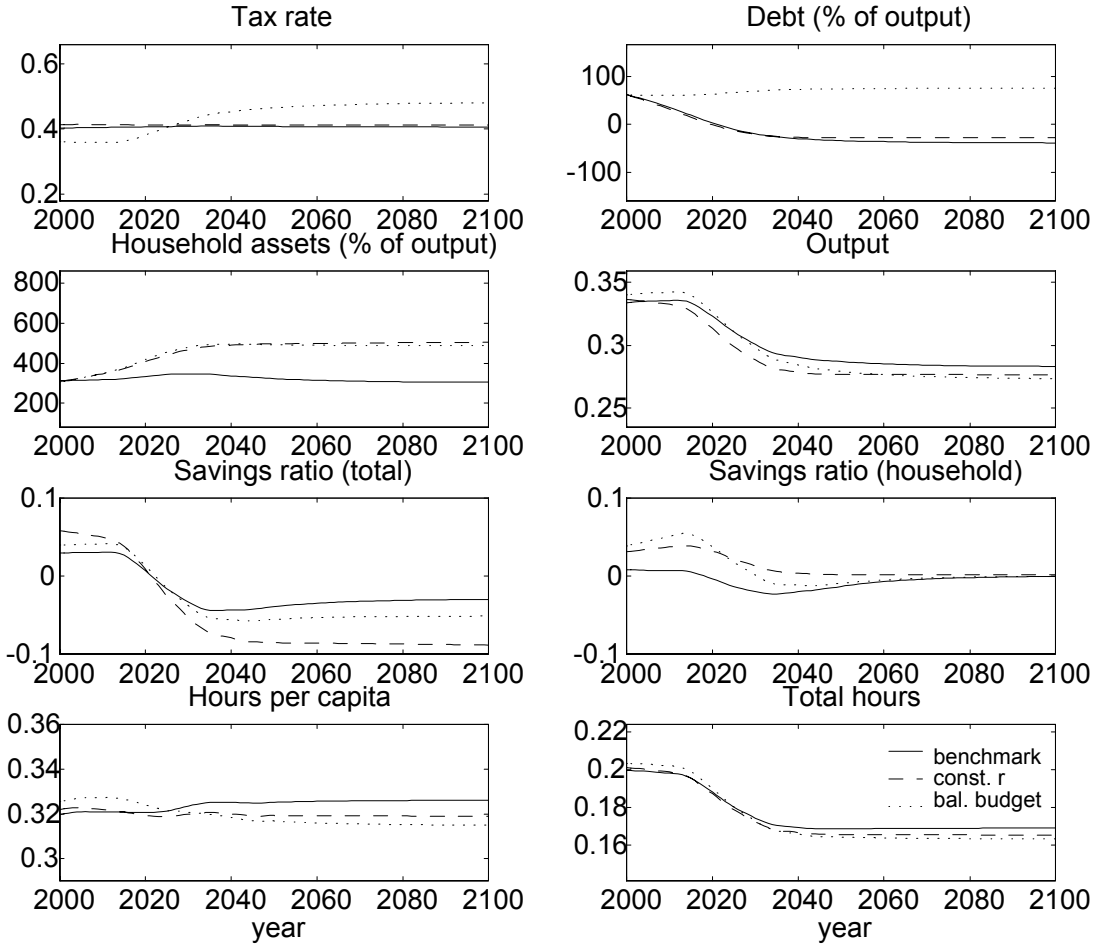


Figure 7E: Optimal policy vs. balanced-budget policy, Germany

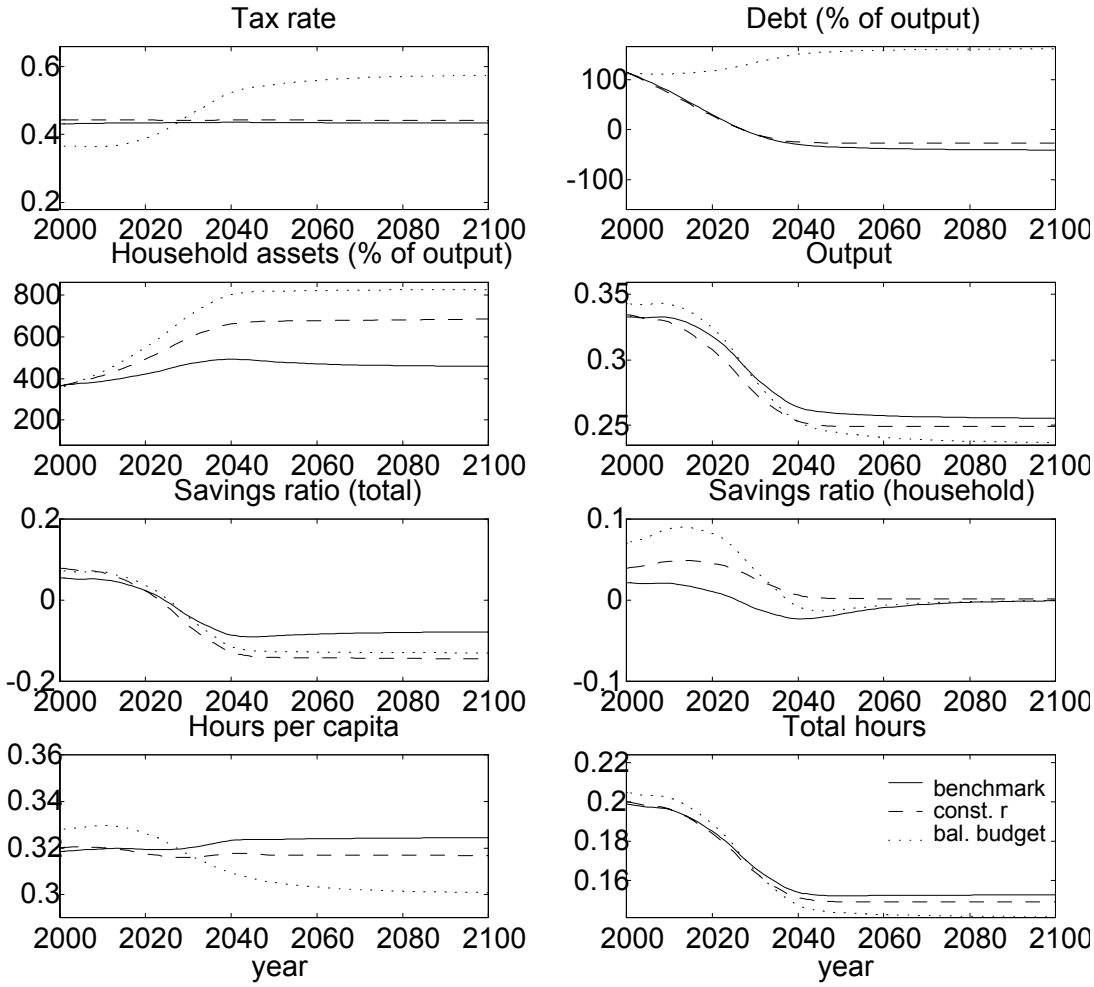


Figure 7F: Optimal policy vs. balanced-budget policy, Italy

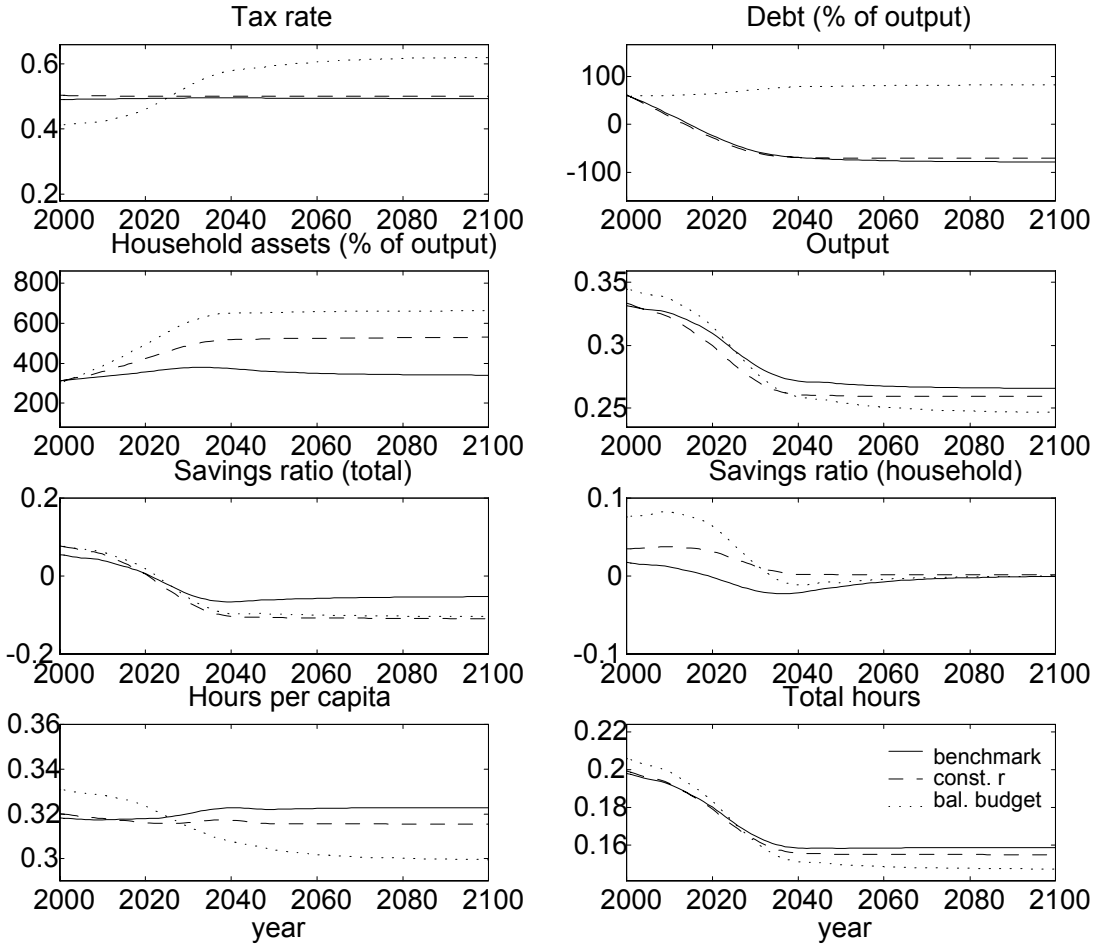


Figure 7G: Optimal policy vs. balanced-budget policy, Netherlands

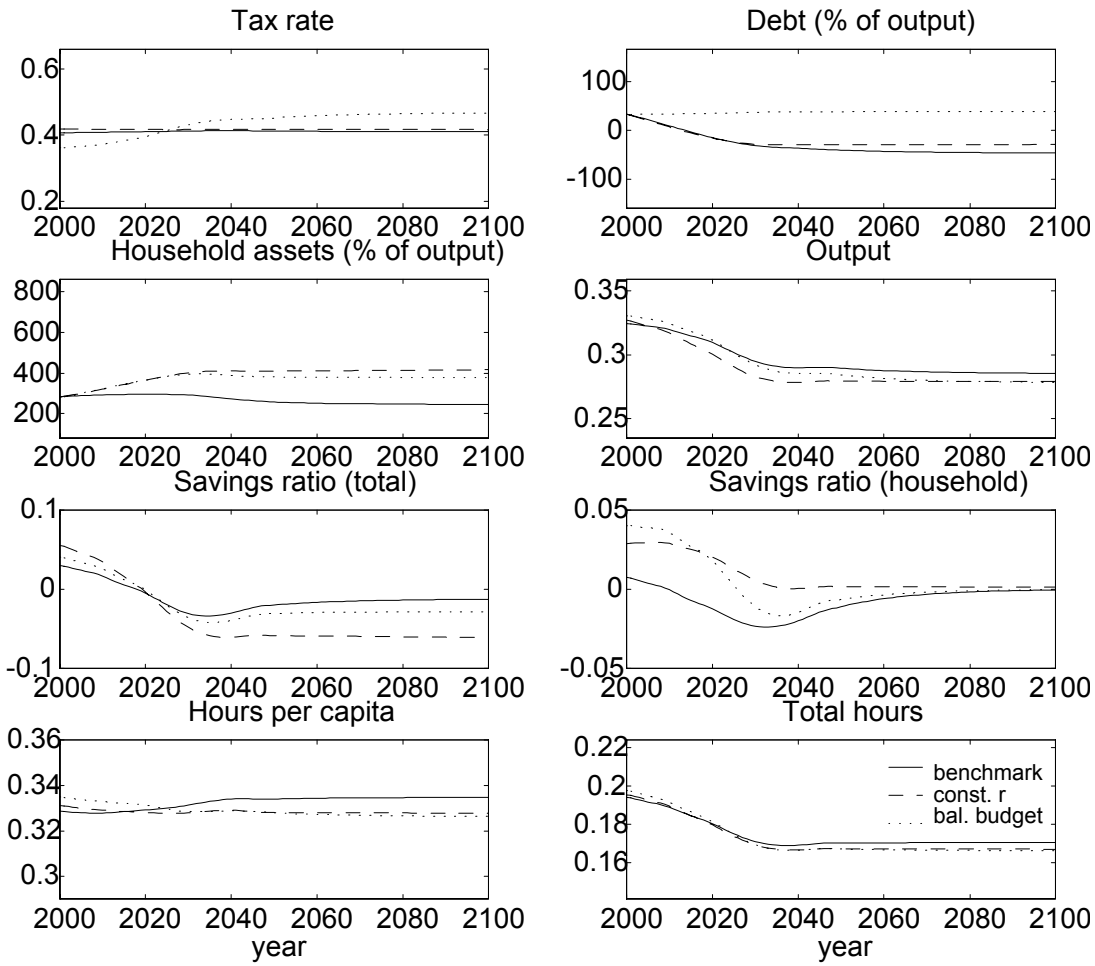


Figure 7H: Optimal policy vs. balanced-budget policy, Norway

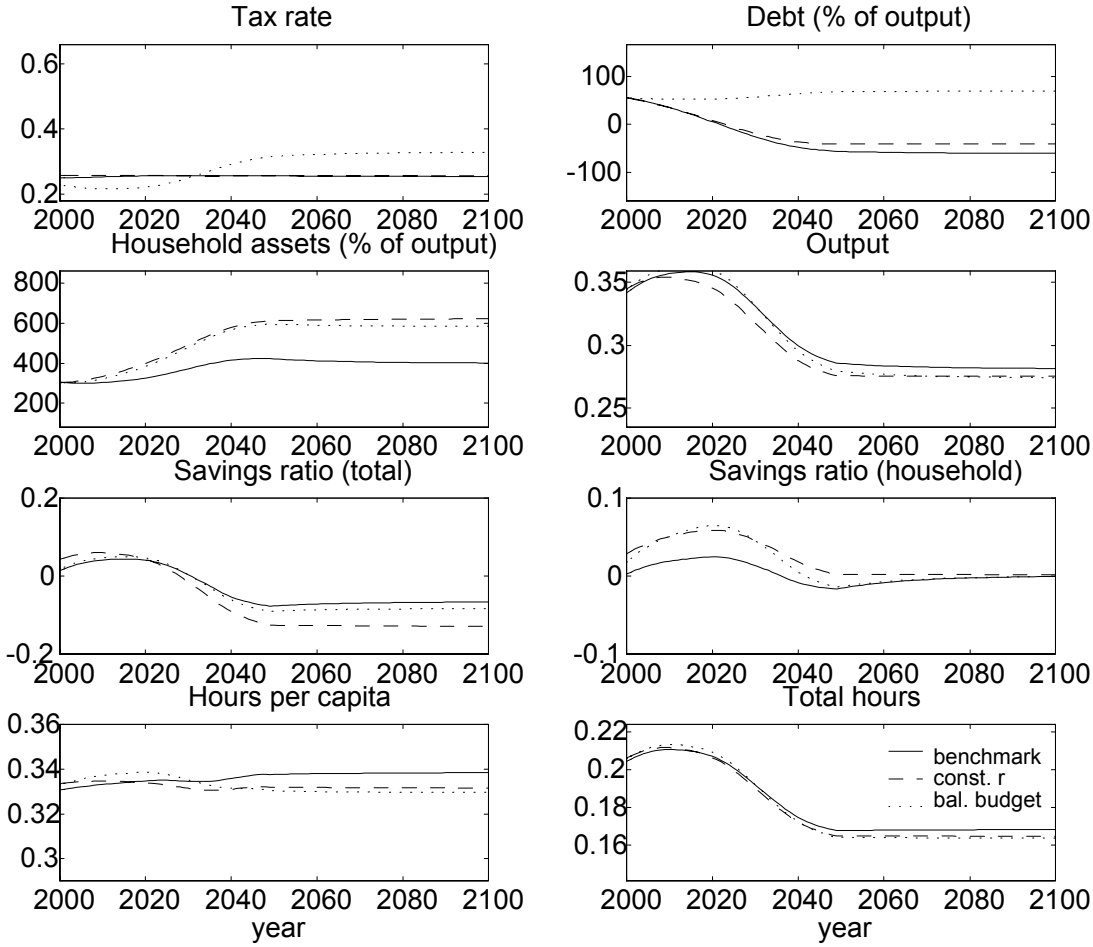


Figure 7I: Optimal policy vs. balanced-budget policy, Portugal

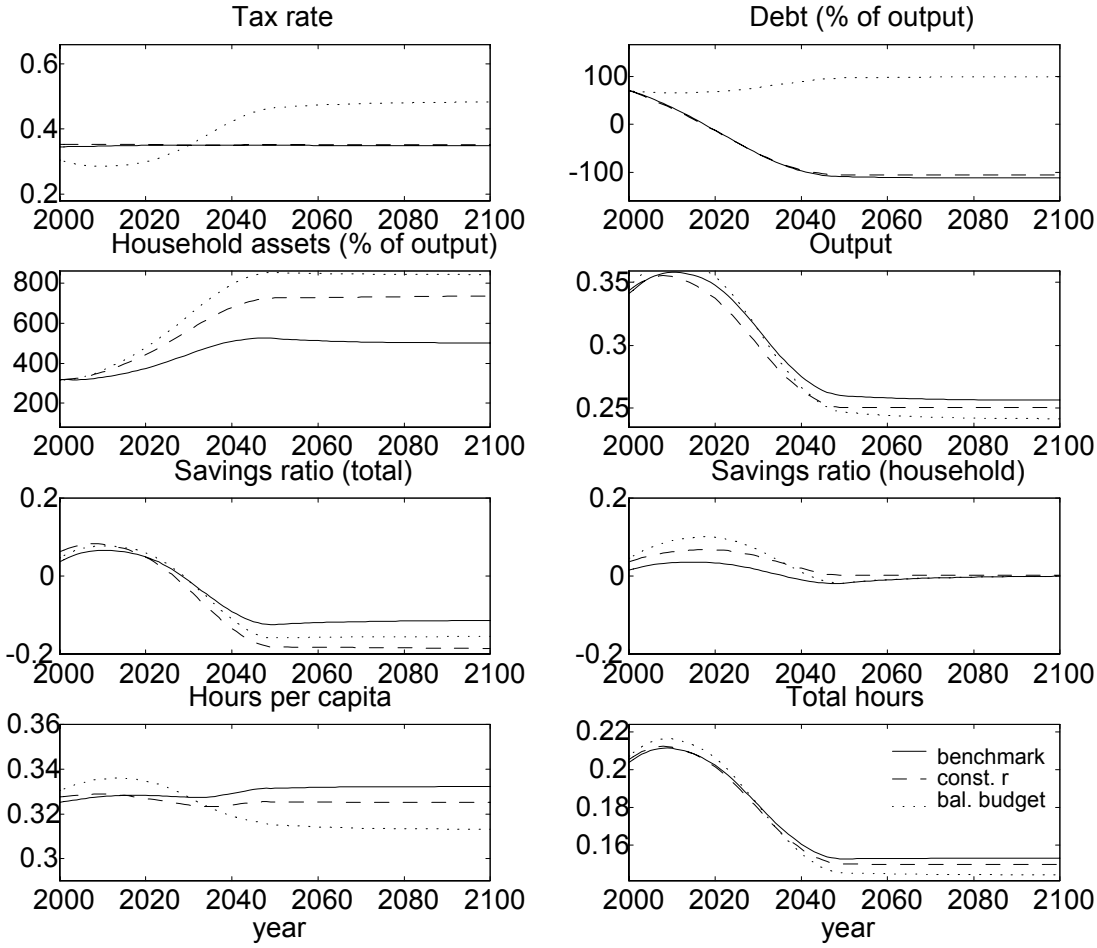


Figure 7J: Optimal policy vs. balanced-budget policy, Spain

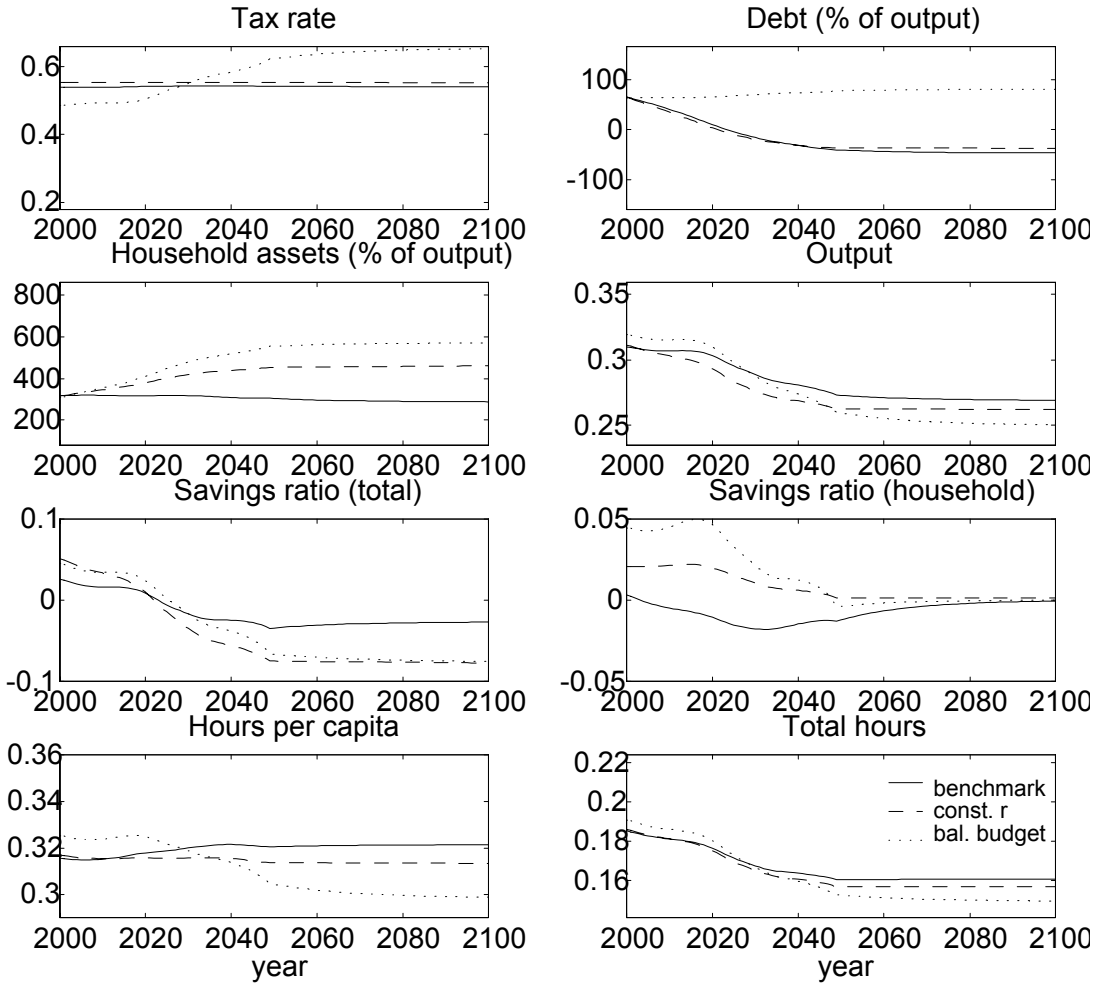


Figure 7K: Optimal policy vs. balanced-budget policy, Sweden

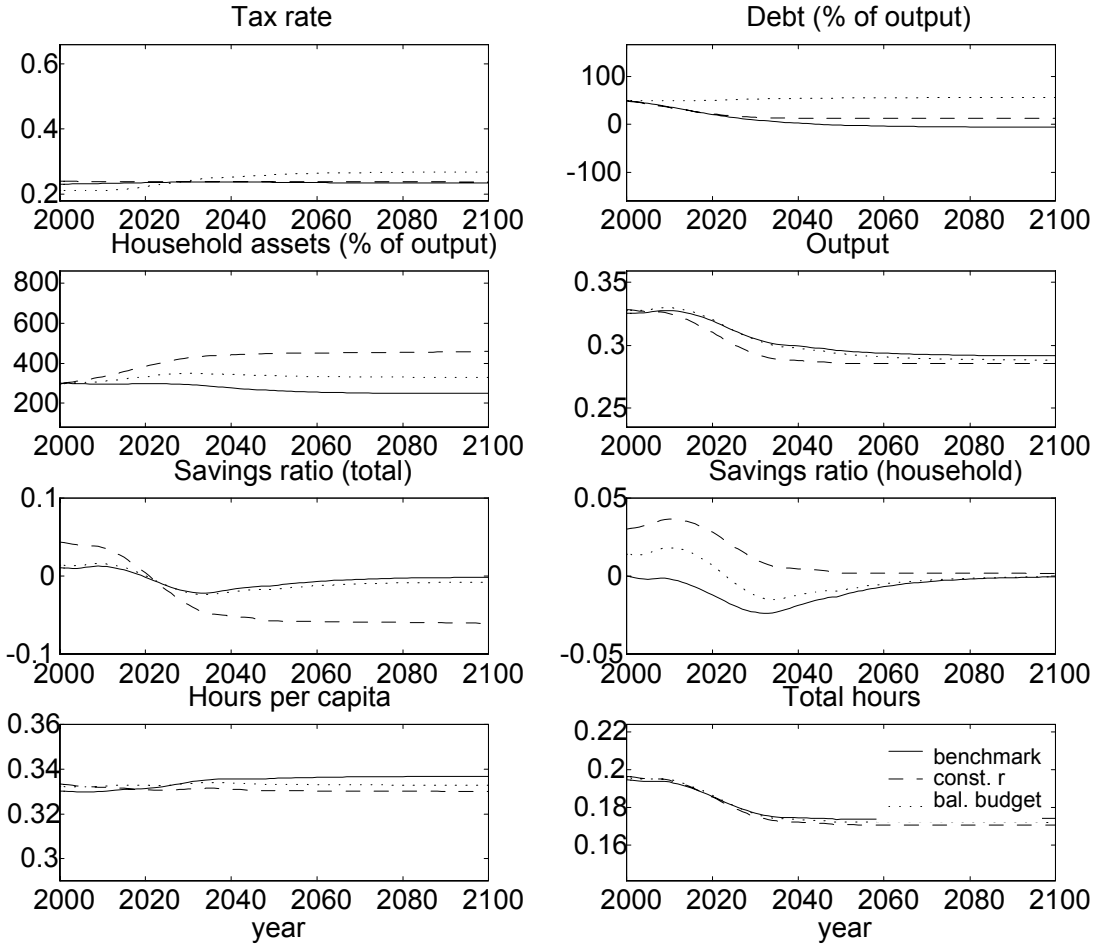


Figure 7L: Optimal policy vs. balanced-budget policy, U.K.

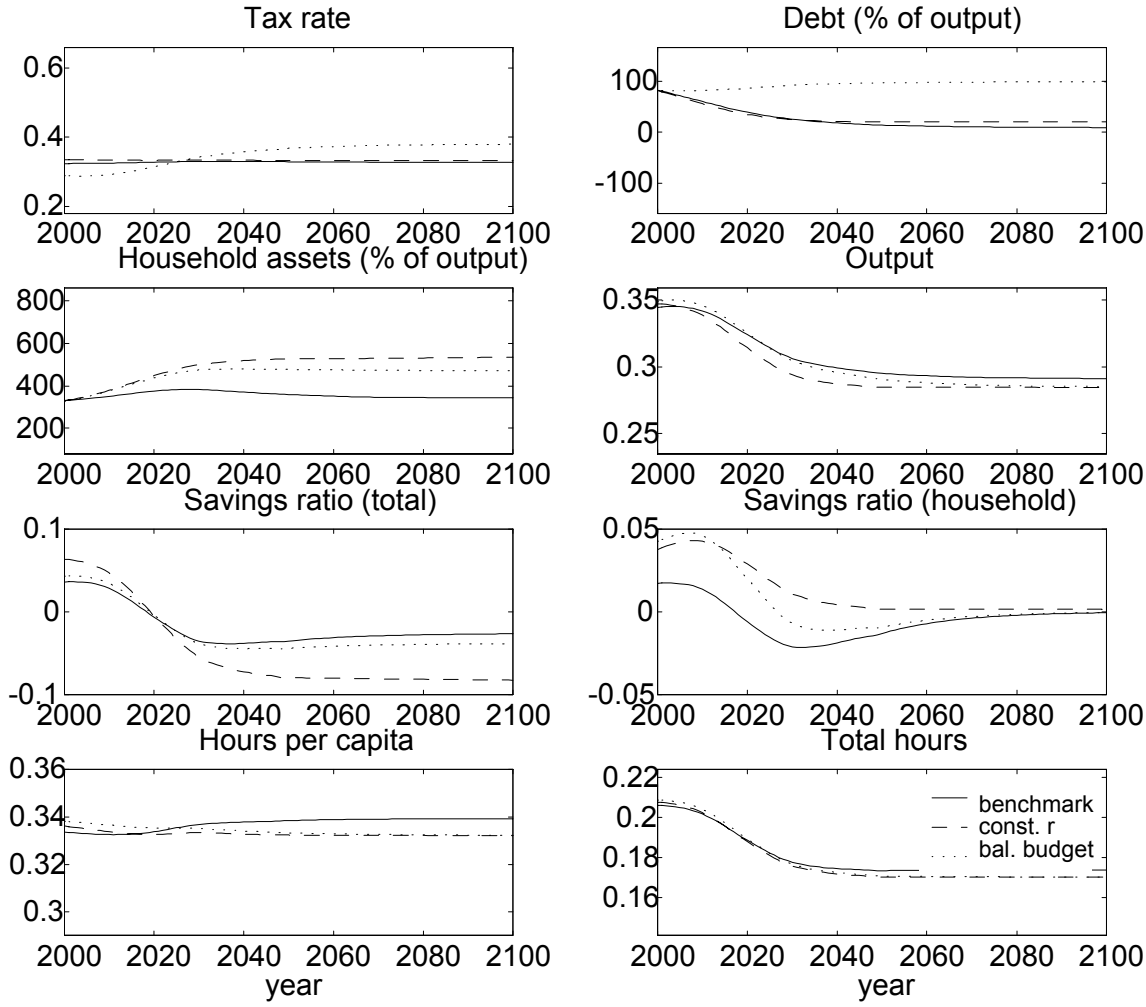


Figure 7M: Optimal policy vs. balanced-budget policy, Canada.

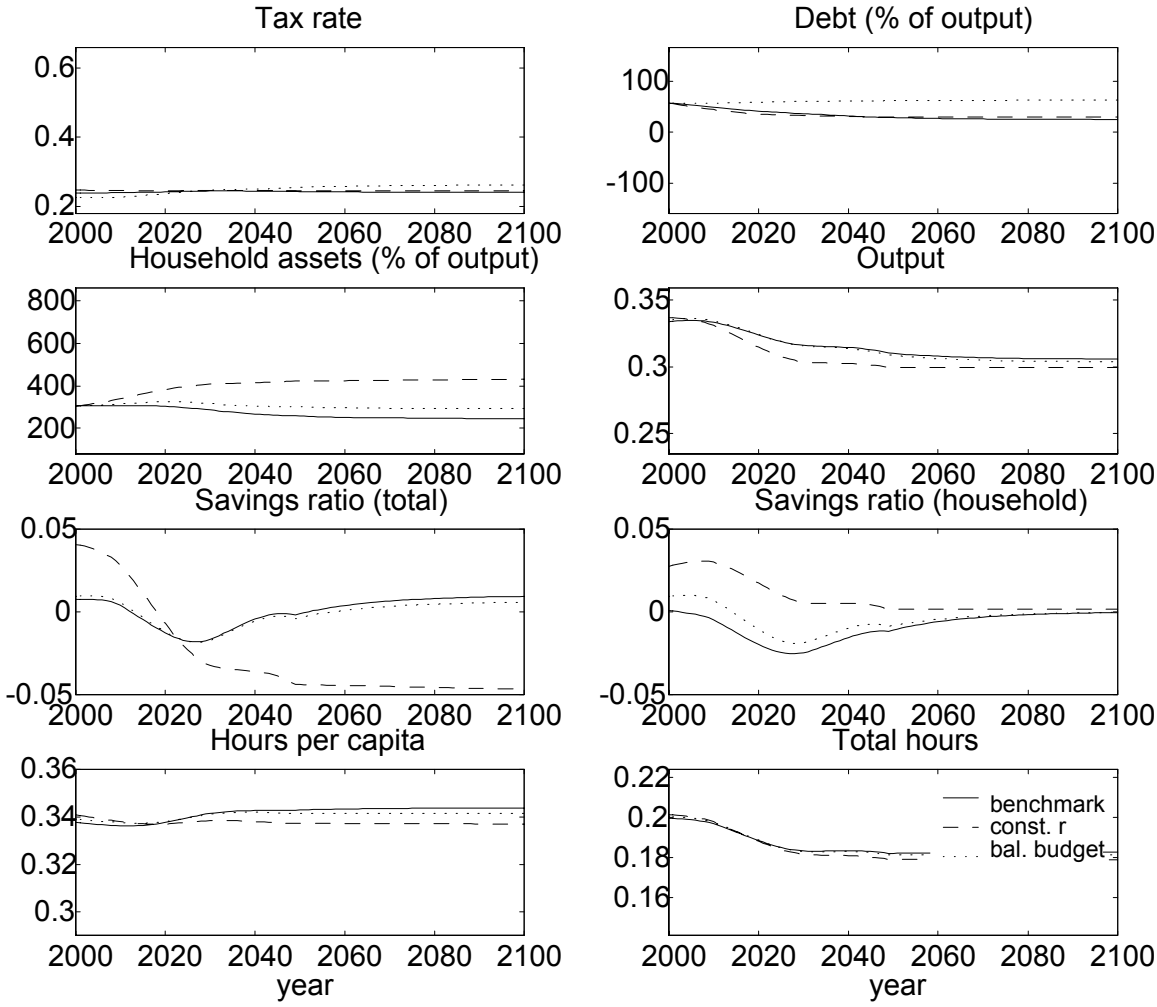


Figure 7N: Optimal policy vs. balanced-budget policy, United States.